

null

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1 Env

```
rm(list = ls())  
setwd("/Users/liqun/Desktop/scMutraceReproducibility/Figure4/")
```

2 Libraries

```
library(FSA)  
## ## FSA v0.10.0. See citation('FSA') if used in publication.  
## ## Run fishR() for related website and fishR('IFAR') for related book.  
library(ineq)  
library(vegan)
```

```

## Loading required package: permute
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
library(tidyr)
library(ggpubr)
## Loading required package: ggplot2
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recommended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following objects are masked from 'package:base':
##
##     intersect, t
library(igraph)
##
## Attaching package: 'igraph'
## The following object is masked from 'package:Seurat':
##
##     components
## The following object is masked from 'package:tidyr':
##
##     crossing
## The following objects are masked from 'package:dplyr':
##
##     as_data_frame, groups, union
## The following object is masked from 'package:vegan':
##
##     diversity
## The following object is masked from 'package:permute':
##
##     permute
## The following objects are masked from 'package:stats':
##
##     decompose, spectrum
## The following object is masked from 'package:base':
##
##     union
library(ggraph)

```

```

##
## Attaching package: 'ggraph'
## The following object is masked from 'package:sp':
##
##      geometry
library(webshot)
library(ggplot2)
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##   method                from
##   heightDetails.titleGrob ggplot2
##   widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##      as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##      annotate
library(stringr)
library(corrplot)
## corrplot 0.95 loaded
library(pheatmap)
library(reshape2)
##
## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##      smiths
library(tidygraph)
##
## Attaching package: 'tidygraph'
## The following object is masked from 'package:igraph':
##
##      groups
## The following object is masked from 'package:stats':
##
##      filter
library(networkD3)
##
## Attaching package: 'networkD3'
## The following object is masked from 'package:Seurat':
##
##      JS
## The following object is masked from 'package:SeuratObject':
##
##      JS
library(tidyverse)
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr 2.1.5
## v lubridate 1.9.4    v tibble 3.3.0

```

```
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%--%()      masks igraph::%--%()
## x ggpp::annotate()      masks ggplot2::annotate()
## x tibble::as_data_frame() masks igraph::as_data_frame(), dplyr::as_data_frame()
## x purrr::compose()      masks igraph::compose()
## x igraph::crossing()     masks tidyr::crossing()
## x tidygraph::filter()   masks dplyr::filter(), stats::filter()
## x dplyr::lag()          masks stats::lag()
## x purrr::simplify()     masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)
```

3 Functions

```
retriveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag",locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[,c("MutNum", "WTNum")]) > 0, ]
  cells_WT <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "WTNum"] > 0,]$CellTag
  cells_Mut <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "MutNum"] > 0,]$CellTag
  cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
  return(cells)
}

retriveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList,]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")]) > 0, ]
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0,]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0,]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}

cal_MultiSpe <- function(matrixData) {
  res <- data.frame(
    Shannon = vegan::diversity(matrixData, index = "shannon"),
    Simpson = vegan::diversity(matrixData, index = "simpson"),
    Gini = apply(matrixData, 1, function(x) ineq(x + 0.00001, type = "Gini")),
  )
}
```

```

    sumMut = rowSums(matrixData),
    Tau = NA
  )

  # Tau = sum(1 - (x / max(x))) / (n - 1)
  res$Tau <- apply(matrixData, 1, function(x) {
    if (all(x == 0)) return(NA)
    x <- x / max(x)
    sum(1 - x) / (length(x) - 1)
  })

  rownames(res) <- rownames(matrixData)
  return(res)
}

scale01 <- function(x) (x - min(x)) / (max(x) - min(x))

cal_scMutScore <- function(data_cell_mut, data_mut_time, data_meta){
  # data_cell_mut [cell * muts]
  # data_mut_time [muts * tissues/celltype/days]
  # data_meta [Cellbarcode, CellMutTag(tissues/celltype/days)]
  # data_cell_time_matrix
  replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

  data_cell_mut_num <- as.data.frame(lapply(data_cell_mut, function(x) {
    ifelse(x %in% names(replacement_map), replacement_map[x], x)
  })))

  rownames(data_cell_mut_num) <- rownames(data_cell_mut)
  print(dim(data_cell_mut_num))
  print(dim(data_mut_time))

  data_cell_time_matrix <- as.matrix(data_cell_mut_num) %*% as.matrix(data_mut_time)
  rownames(data_cell_time_matrix) <- rownames(data_cell_mut)
  colnames(data_cell_time_matrix) <- colnames(data_mut_time)

  muts_total <- unique(rownames(data_mut_time))

  # get different tissues/time/celltype names
  col_names_data <- colnames(data_cell_time_matrix)

  col_length <- length(col_names_data)

  col_coefficient_fix <- setNames(
    rep(1/col_length, col_length),
    col_names_data
  )

  data_cell_time_matrix <- data_cell_time_matrix[,col_names_data]

  # Gini
  res_index <- cal_MultiSpe(data_cell_time_matrix)

```

```

res_diversity <- as.data.frame((data_cell_time_matrix > 1) * rep(col_coefficient_fix[colnames(data_cell_time_matrix)], ncol(data_cell_time_matrix)))
res_diversity$Diversity <- rowSums(res_diversity)

res_diversity$Diversity[is.na(res_diversity$Diversity)] <- 0.00001

res_index$Gini[is.na(res_index$Gini)] <- 0.00001
res_index$Tau[is.na(res_index$Tau)] <- 0.00001

res_index$GiniTau <- res_index$Gini * res_index$Tau

res_index$scMutScore <- 2 / (
  (1 / (0.1 * res_index$GiniTau + 0.00001)) +
  (1 / (0.9 * res_diversity$Diversity + 0.00001))
)

res_index$Diversity <- res_diversity$Diversity

#res_index$scMutScoreQun <- 2 / (
#  (1 / (0.2 * res_index$sumMut + 0.00001)) +
#  (1 / (0.8 * res_index$GiniTau + 0.00001)) +
#  (1 / (0.9 * res_diversity$Diversity + 0.00001))
#)

#res_index$scMutScore <- 2 / (
#  (1 / (0.1 * scale01(res_index$sumMut) + 0.00001)) +
#  (1 / (0.9 * res_diversity$Diversity + 0.00001))
#)

res_index$scMutScoreQun <- res_index$sumMut

res_index$scMutScore <- scales::rescale(res_index$scMutScore, to = c(0, 1))
res_index$scMutScoreQun <- scales::rescale(res_index$scMutScoreQun, to = c(0, 1))

res <- list(diversity = res_diversity, score = res_index, matrixD = data_cell_time_matrix)
return(res)
}

```

4 LoadData

```

CD34_CSM <- readRDS("../Data/CD34/CD34_CSM.rds")
CD34_SSM <- readRDS("../Data/CD34/CD34_SSM.rds")
CD34_Meta <- readRDS("../Data/CD34/CD34_meta.rds")

```

5 Calculationg mutations of each cell and cells of each mutation

```

CD34_MutPerCell <- data.frame(
  CellTag = rownames(CD34_CSM),
  MutNum_somatic = rowSums(CD34_CSM == "Mut"),

```

```

  WTNum_somatic = rowSums(CD34_CSM == "WT")
)

CD34_CellPerMut <- data.frame(
  Location = colnames(CD34_CSM),
  CellMutNum = colSums(CD34_CSM == "Mut"),
  CellWTNum = colSums(CD34_CSM == "WT")
)

cells_candidate <- CD34_MutPerCell[CD34_MutPerCell$MutNum_somatic >= 1,]$CellTag
mut_candidate <- CD34_CellPerMut[CD34_CellPerMut$CellMutNum >= 1,]$Location
CD34_CSM <- CD34_CSM[cells_candidate, mut_candidate]

```

6 calMulti

```

time_types <- c("ery1", "ery2", "ery3", "ery4", "ery5", "ery6", "my1", "my2", "my3", "my4", "prog",
               "ery7", "ery8", "ery9", "ery10", "ery11", "ery12", "ery13", "ery14", "ery15", "ery16", "ery17", "ery18", "ery19", "ery20", "ery21", "ery22", "ery23", "ery24", "ery25", "ery26", "ery27", "ery28", "ery29", "ery30", "ery31", "ery32", "ery33", "ery34", "ery35", "ery36", "ery37", "ery38", "ery39", "ery40", "ery41", "ery42", "ery43", "ery44", "ery45", "ery46", "ery47", "ery48", "ery49", "ery50", "ery51", "ery52", "ery53", "ery54", "ery55", "ery56", "ery57", "ery58", "ery59", "ery60", "ery61", "ery62", "ery63", "ery64", "ery65", "ery66", "ery67", "ery68", "ery69", "ery70", "ery71", "ery72", "ery73", "ery74", "ery75", "ery76", "ery77", "ery78", "ery79", "ery80", "ery81", "ery82", "ery83", "ery84", "ery85", "ery86", "ery87", "ery88", "ery89", "ery90", "ery91", "ery92", "ery93", "ery94", "ery95", "ery96", "ery97", "ery98", "ery99", "ery100")

# Cell_Mut
CD34_Meta <- CD34_Meta[CD34_Meta$Cellbarcode %in% cells_candidate,]
CD34_cell_mut_raw <- CD34_CSM

# Mut_Time
CD34_mut_time <- as.data.frame(matrix(nrow = length(mut_candidate), ncol = length(time_types)))
rownames(CD34_mut_time) <- mut_candidate
colnames(CD34_mut_time) <- time_types

for (i in 1:nrow(CD34_mut_time)){
  mut_tmp <- rownames(CD34_mut_time)[i]
  cell_tmp <- retrieveCellsByLocation(CD34_cell_mut_raw, mut_tmp)
  selected_meta_tags <- CD34_Meta[CD34_Meta$Cellbarcode %in% cell_tmp$Mut,]$Group

  CD34_mut_time[i,]$ery1 <- sum(str_count(selected_meta_tags, "ery1"))
  CD34_mut_time[i,]$ery2 <- sum(str_count(selected_meta_tags, "ery2"))
  CD34_mut_time[i,]$ery3 <- sum(str_count(selected_meta_tags, "ery3"))

  CD34_mut_time[i,]$ery4 <- sum(str_count(selected_meta_tags, "ery4"))
  CD34_mut_time[i,]$ery5 <- sum(str_count(selected_meta_tags, "ery5"))
  CD34_mut_time[i,]$ery6 <- sum(str_count(selected_meta_tags, "ery6"))

  CD34_mut_time[i,]$my1 <- sum(str_count(selected_meta_tags, "my1"))
  CD34_mut_time[i,]$my2 <- sum(str_count(selected_meta_tags, "my2"))
  CD34_mut_time[i,]$my3 <- sum(str_count(selected_meta_tags, "my3"))
  CD34_mut_time[i,]$my4 <- sum(str_count(selected_meta_tags, "my4"))

  CD34_mut_time[i,]$prog <- sum(str_count(selected_meta_tags, "prog"))
  CD34_mut_time[i,]$prog_my <- sum(str_count(selected_meta_tags, "prog_my"))
  CD34_mut_time[i,]$prog_ery <- sum(str_count(selected_meta_tags, "prog_ery"))
}

```

```

res_index_raw_total <- cal_scMutScore(CD34_cell_mut_raw, CD34_mut_time, CD34_Meta)
## [1] 1893 478
## [1] 478 13

res_index_raw <- res_index_raw_total$score
res_index_raw$Cellbarcode <- rownames(res_index_raw)

res_matrix <- res_index_raw_total$matrixD

res_index <- merge(res_index_raw, CD34_Meta, by = "Cellbarcode")

selectedGroup <- c("prog", "prog_my", "prog_ery", "my1", "my2", "my3", "my4", "ery1", "ery2", "ery3", "ery4", "ery5", "ery6", "ery7", "ery8", "ery9", "ery10", "ery11", "ery12", "ery13", "ery14", "ery15", "ery16", "ery17", "ery18", "ery19", "ery20", "ery21", "ery22", "ery23", "ery24", "ery25", "ery26", "ery27", "ery28", "ery29", "ery30", "ery31", "ery32", "ery33", "ery34", "ery35", "ery36", "ery37", "ery38", "ery39", "ery40", "ery41", "ery42", "ery43", "ery44", "ery45", "ery46", "ery47", "ery48", "ery49", "ery50", "ery51", "ery52", "ery53", "ery54", "ery55", "ery56", "ery57", "ery58", "ery59", "ery60", "ery61", "ery62", "ery63", "ery64", "ery65", "ery66", "ery67", "ery68", "ery69", "ery70", "ery71", "ery72", "ery73", "ery74", "ery75", "ery76", "ery77", "ery78", "ery79", "ery80", "ery81", "ery82", "ery83", "ery84", "ery85", "ery86", "ery87", "ery88", "ery89", "ery90", "ery91", "ery92", "ery93", "ery94", "ery95", "ery96", "ery97", "ery98", "ery99", "ery100")
res_index <- res_index[res_index$Group %in% selectedGroup,]

#res_index$Group[grepl('prog/prog_my/prog_ery/mix',res_index$Group)] <- 'Prog'

res_index$CellMutTag <- factor(res_index$CellMutTag, levels = c("D_8", "D_14", "D_20"))

cluster_color <- c(my1 = '#6baed6', my2 = '#4292c6', my3 = '#2171b5', my4 = '#084594', prog = '#e0e0e0',
  ery1 = "#bfd3e6", ery2 = "#9ebcda", ery3 = "#8c96c6", ery4 = "#8c6bb1", ery5 = "#884d87", ery6 = "#8c564b",
  ery7 = "#e377c2", ery8 = "#17becf", ery9 = "#1f77b4", ery10 = "#1f77b4", ery11 = "#1f77b4", ery12 = "#1f77b4",
  ery13 = "#1f77b4", ery14 = "#1f77b4", ery15 = "#1f77b4", ery16 = "#1f77b4", ery17 = "#1f77b4", ery18 = "#1f77b4",
  ery19 = "#1f77b4", ery20 = "#1f77b4", ery21 = "#1f77b4", ery22 = "#1f77b4", ery23 = "#1f77b4", ery24 = "#1f77b4",
  ery25 = "#1f77b4", ery26 = "#1f77b4", ery27 = "#1f77b4", ery28 = "#1f77b4", ery29 = "#1f77b4", ery30 = "#1f77b4",
  ery31 = "#1f77b4", ery32 = "#1f77b4", ery33 = "#1f77b4", ery34 = "#1f77b4", ery35 = "#1f77b4", ery36 = "#1f77b4",
  ery37 = "#1f77b4", ery38 = "#1f77b4", ery39 = "#1f77b4", ery40 = "#1f77b4", ery41 = "#1f77b4", ery42 = "#1f77b4",
  ery43 = "#1f77b4", ery44 = "#1f77b4", ery45 = "#1f77b4", ery46 = "#1f77b4", ery47 = "#1f77b4", ery48 = "#1f77b4",
  ery49 = "#1f77b4", ery50 = "#1f77b4", ery51 = "#1f77b4", ery52 = "#1f77b4", ery53 = "#1f77b4", ery54 = "#1f77b4",
  ery55 = "#1f77b4", ery56 = "#1f77b4", ery57 = "#1f77b4", ery58 = "#1f77b4", ery59 = "#1f77b4", ery60 = "#1f77b4",
  ery61 = "#1f77b4", ery62 = "#1f77b4", ery63 = "#1f77b4", ery64 = "#1f77b4", ery65 = "#1f77b4", ery66 = "#1f77b4",
  ery67 = "#1f77b4", ery68 = "#1f77b4", ery69 = "#1f77b4", ery70 = "#1f77b4", ery71 = "#1f77b4", ery72 = "#1f77b4",
  ery73 = "#1f77b4", ery74 = "#1f77b4", ery75 = "#1f77b4", ery76 = "#1f77b4", ery77 = "#1f77b4", ery78 = "#1f77b4",
  ery79 = "#1f77b4", ery80 = "#1f77b4", ery81 = "#1f77b4", ery82 = "#1f77b4", ery83 = "#1f77b4", ery84 = "#1f77b4",
  ery85 = "#1f77b4", ery86 = "#1f77b4", ery87 = "#1f77b4", ery88 = "#1f77b4", ery89 = "#1f77b4", ery90 = "#1f77b4",
  ery91 = "#1f77b4", ery92 = "#1f77b4", ery93 = "#1f77b4", ery94 = "#1f77b4", ery95 = "#1f77b4", ery96 = "#1f77b4",
  ery97 = "#1f77b4", ery98 = "#1f77b4", ery99 = "#1f77b4", ery100 = "#1f77b4")
prog_ery = "#d8daeb", mix = "#66c2a4", Prog = '#c6dbef', ery1_2 = "#8c96c6", ery3_4 = "#8c96c6", ery5_6 = "#8c96c6",
  ery7_8 = "#8c96c6", ery9_10 = "#8c96c6", ery11_12 = "#8c96c6", ery13_14 = "#8c96c6", ery15_16 = "#8c96c6",
  ery17_18 = "#8c96c6", ery19_20 = "#8c96c6", ery21_22 = "#8c96c6", ery23_24 = "#8c96c6", ery25_26 = "#8c96c6",
  ery27_28 = "#8c96c6", ery29_30 = "#8c96c6", ery31_32 = "#8c96c6", ery33_34 = "#8c96c6", ery35_36 = "#8c96c6",
  ery37_38 = "#8c96c6", ery39_40 = "#8c96c6", ery41_42 = "#8c96c6", ery43_44 = "#8c96c6", ery45_46 = "#8c96c6",
  ery47_48 = "#8c96c6", ery49_50 = "#8c96c6", ery51_52 = "#8c96c6", ery53_54 = "#8c96c6", ery55_56 = "#8c96c6",
  ery57_58 = "#8c96c6", ery59_60 = "#8c96c6", ery61_62 = "#8c96c6", ery63_64 = "#8c96c6", ery65_66 = "#8c96c6",
  ery67_68 = "#8c96c6", ery69_70 = "#8c96c6", ery71_72 = "#8c96c6", ery73_74 = "#8c96c6", ery75_76 = "#8c96c6",
  ery77_78 = "#8c96c6", ery79_80 = "#8c96c6", ery81_82 = "#8c96c6", ery83_84 = "#8c96c6", ery85_86 = "#8c96c6",
  ery87_88 = "#8c96c6", ery89_90 = "#8c96c6", ery91_92 = "#8c96c6", ery93_94 = "#8c96c6", ery95_96 = "#8c96c6",
  ery97_98 = "#8c96c6", ery99_100 = "#8c96c6")

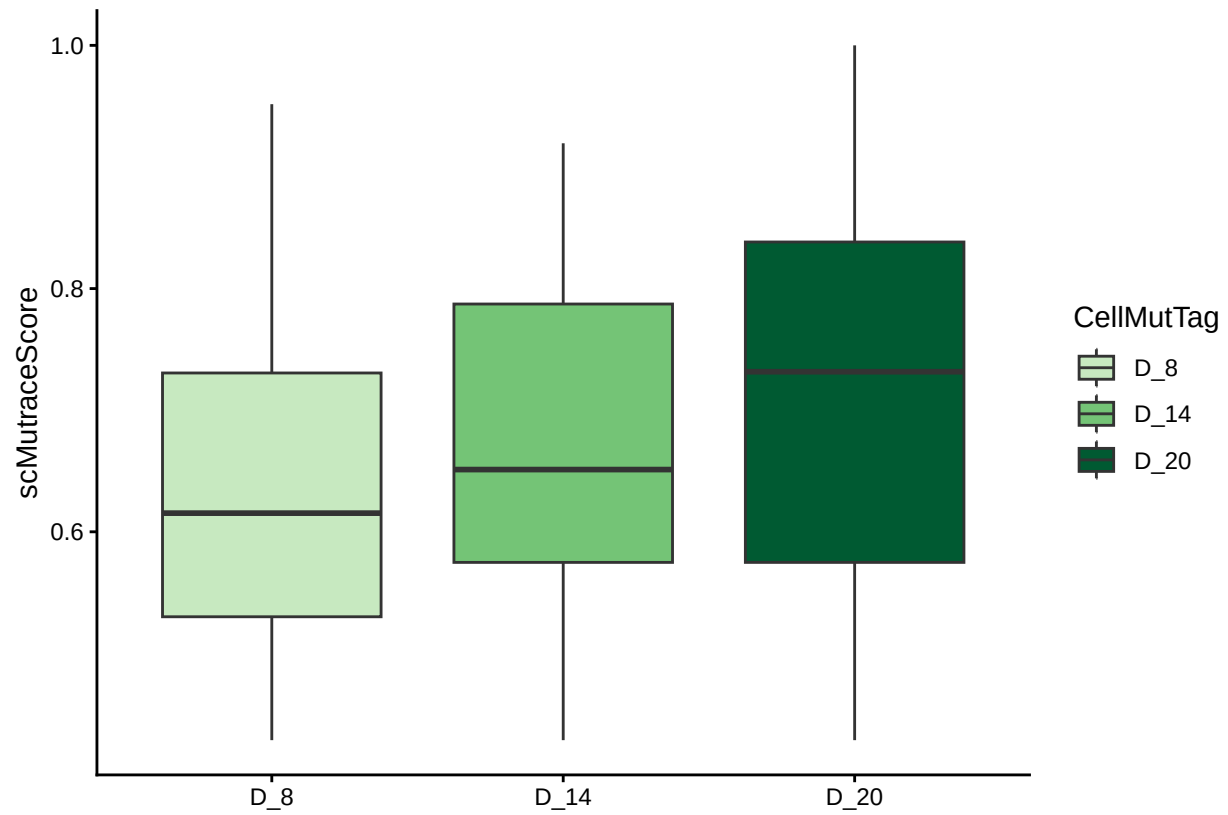
day_color <- c(D_8 = "#c7e9c0", D_14 = "#74c476", D_20 = "#005a32")

cell_group_color <- c(Prog = "#e5f5f9", Diff = "#66c2a4", Terminal = "#006d2c")

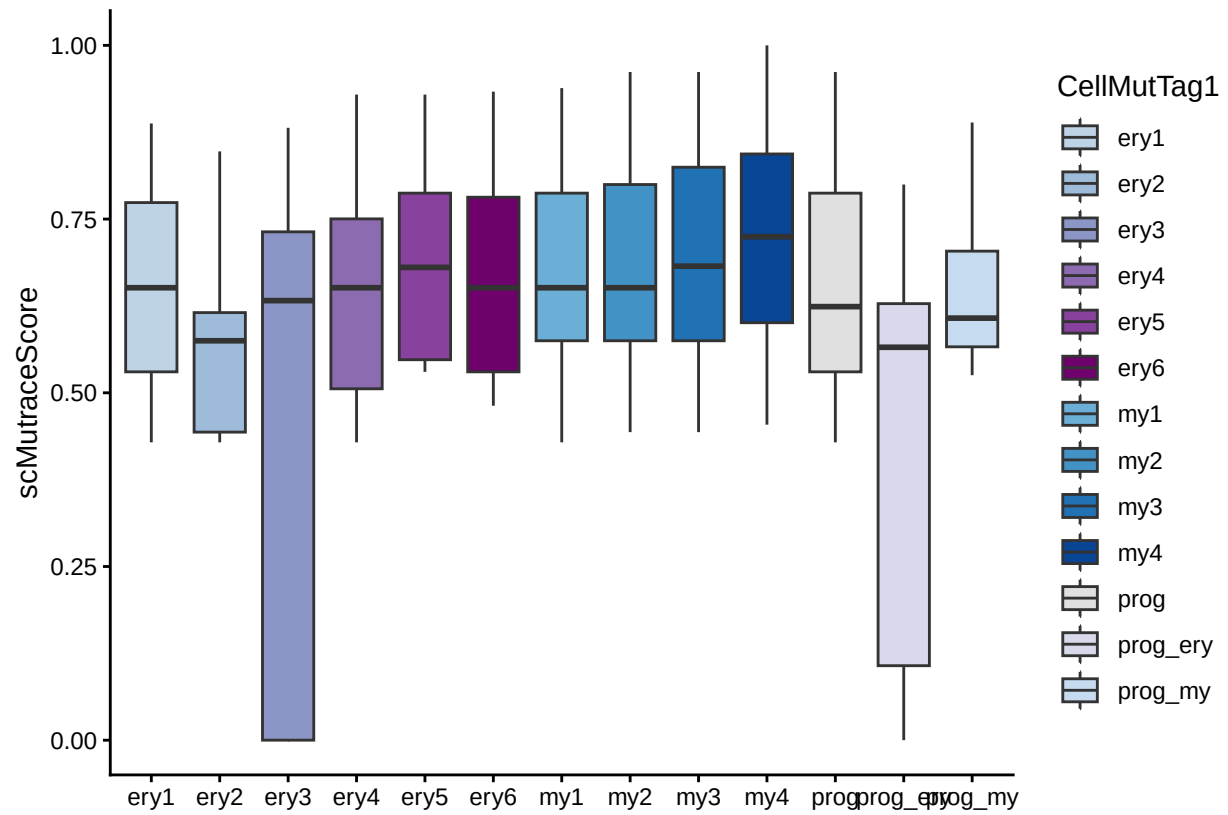
res_index <- res_index[!is.na(res_index$scMutScore),]

ggplot(data = res_index, mapping = aes(x = CellMutTag, y = scMutScore, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()

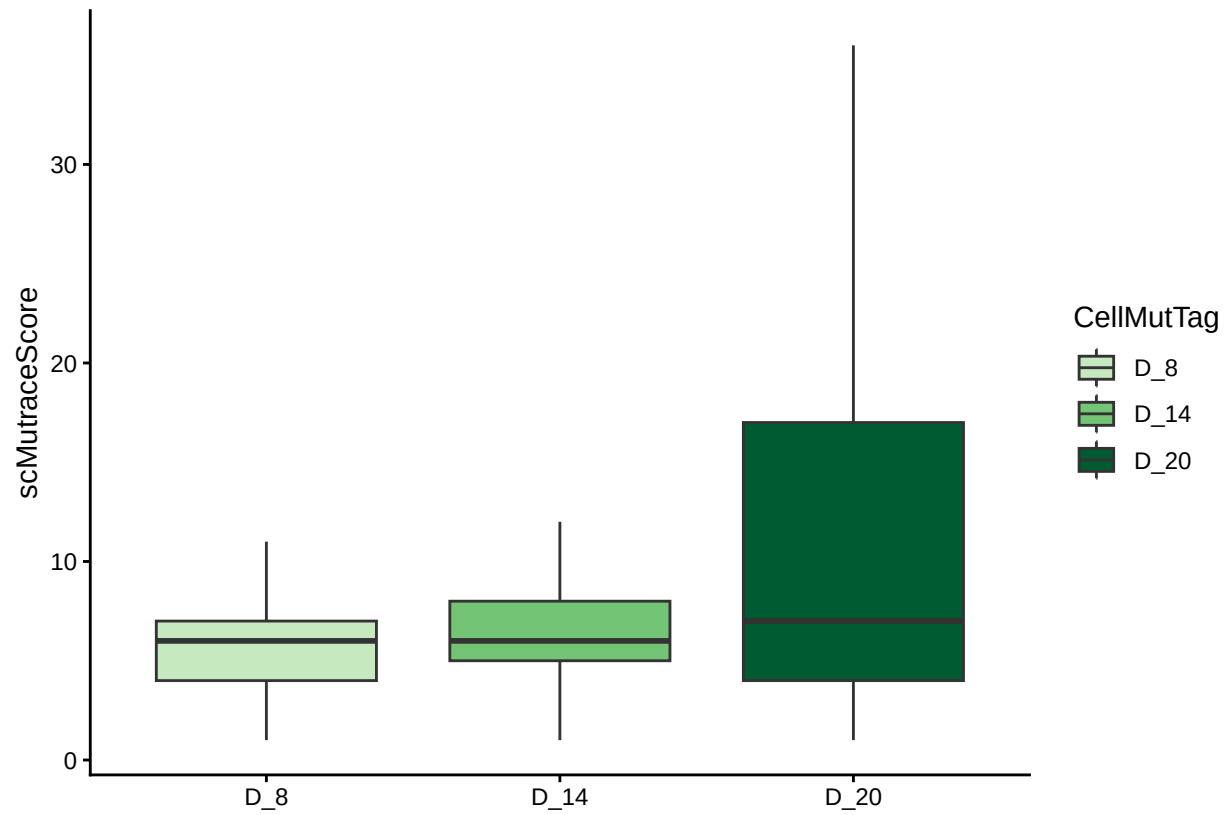
```

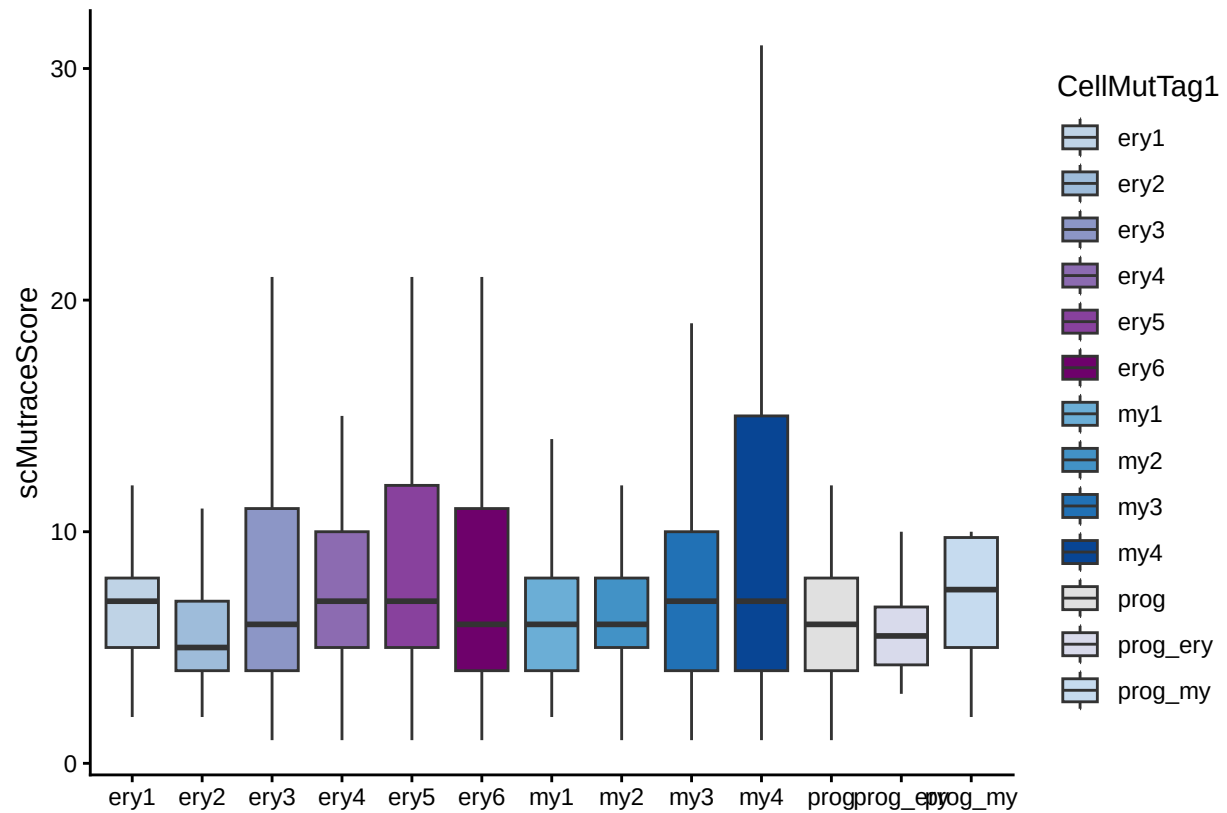
```
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = scMutScore, fill = CellMutTag1)) +  
  geom_boxplot(outliers = FALSE) +  
  labs(x = "", y = "scMutraceScore") +  
  scale_fill_manual(values = cluster_color) +  
  theme_classic()
```



```
ggplot(data = res_index, mapping = aes(x = CellMutTag, y = sumMut, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()
```

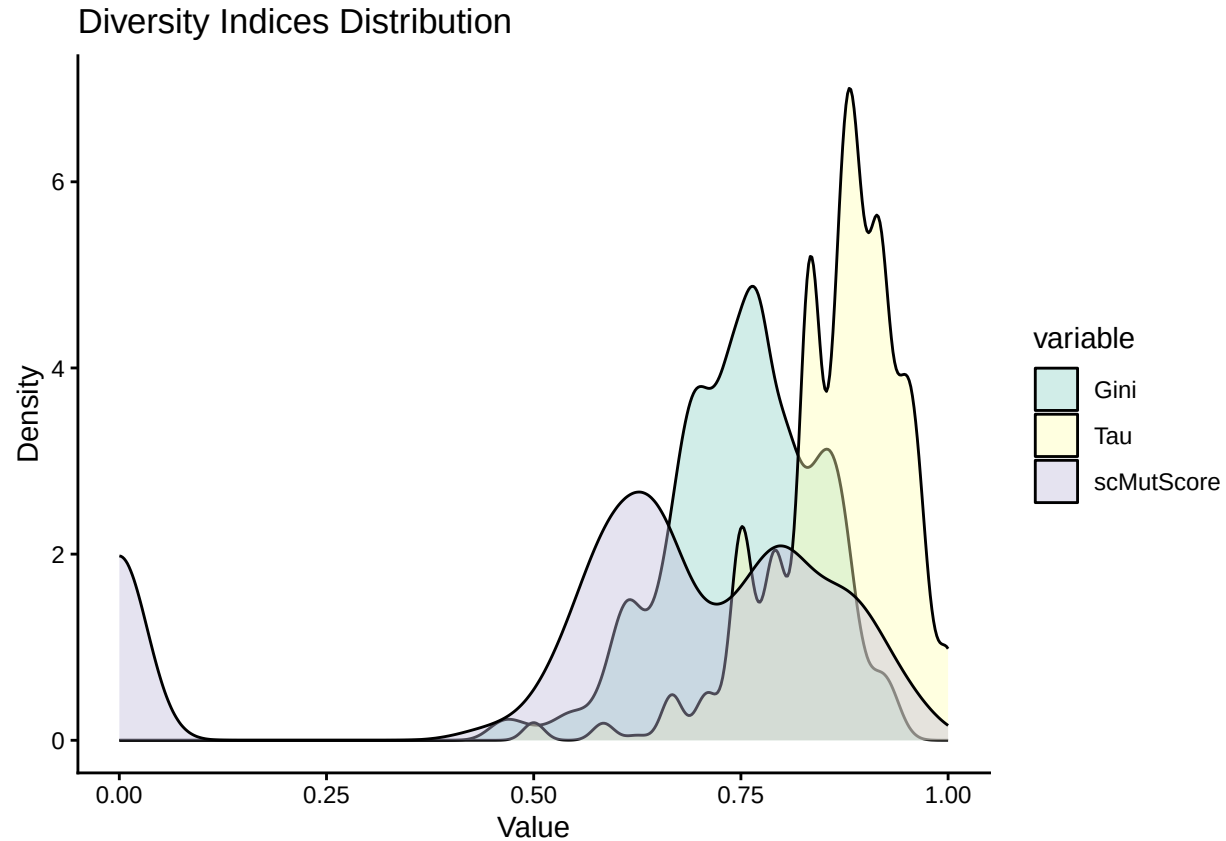


```
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = sumMut, fill = CellMutTag1)) +  
  geom_boxplot(outliers = FALSE) +  
  labs(x = "", y = "scMutraceScore") +  
  scale_fill_manual(values = cluster_color) +  
  theme_classic()
```



```
res_index_long <- melt(res_index[,c("Gini", "Tau", "scMutScore")])
## No id variables; using all as measure variables

ggplot(res_index_long, aes(x=value, fill=variable)) +
  geom_density(alpha=0.4) +
  theme_classic() +
  labs(x="Value", y="Density", title="Diversity Indices Distribution") +
  scale_fill_brewer(palette="Set3")
```



7 outputFigures

```
res_index <- res_index[res_index$Diversity > 0, ]

res_index %>% dplyr::filter(Group %in% c("Prog", "prog_my", "prog_ery", "prog", "my1", "my2", "my3", "my4"))
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `groups` argument.
res_index_meta_summary_scMutScore$CultureDay <- paste("D_", res_index_meta_summary_scMutScore$cell_day, sep="")

res_index_meta_summary_scMutScore$cell_group <- 'Prog'
res_index_meta_summary_scMutScore$cell_group[grepl('my1', res_index_meta_summary_scMutScore$Group)] <- 'my1'
res_index_meta_summary_scMutScore$cell_group[grepl('my2|my3|my4', res_index_meta_summary_scMutScore$Group)] <- 'my2'
res_index_meta_summary_scMutScore$cell_group <- factor(res_index_meta_summary_scMutScore$cell_group, levels=c("Prog", "my1", "my2"))

res_index$GroupQun <- as.character(res_index$Group)

res_index$GroupQun[grepl("ery1|ery2", res_index$Group)] <- "ery1_2"
res_index$GroupQun[grepl("ery3|ery4", res_index$Group)] <- "ery3_4"
res_index$GroupQun[grepl("ery5|ery6", res_index$Group)] <- "ery5_6"

res_index$Group <- factor(res_index$Group, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my3", "my4"))
res_index$GroupQun <- factor(res_index$GroupQun, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my3", "my4"))
res_index_meta_summary_scMutScore$CultureDay <- factor(res_index_meta_summary_scMutScore$CultureDay, levels=c("D_1", "D_2", "D_3", "D_4", "D_5", "D_6"))
```

```

CD34_Meta_selected <- CD34_Meta[CD34_Meta$Cellbarcode %in% res_index$Cellbarcode,]
CD34_Meta_selected$GroupQun <- CD34_Meta_selected$Group
CD34_Meta_selected$GroupQun[grepl("ery1|ery2", CD34_Meta_selected$Group)] <- "ery1_2"
CD34_Meta_selected$GroupQun[grepl("ery3|ery4", CD34_Meta_selected$Group)] <- "ery3_4"
CD34_Meta_selected$GroupQun[grepl("ery5|ery6", CD34_Meta_selected$Group)] <- "ery5_6"

UMAP_Day8 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 8),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day14 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 14),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day20 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 20),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_Days <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = CellMutTag)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = day_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_Groups <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = GroupQun)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_pseudotime_my <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = pseudotime)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_pseudotime_ery <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = pseudotime)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +

```

```

xlab("") + ylab("") +
theme_classic()

# plot scMutScore by days
p_scMutScorebydays <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=scMutScore, fill = CellMutTag)) +
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

# plot scMutScore by cell types
p_scMutScorebycelltypes <- ggplot(res_index, aes(x=as.factor(Group), y=scMutScore, fill = Group)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_scMutScorebycelltypesQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=scMutScore, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  ylim(0,1) +
  theme_classic()

p_EpiScorebyCelltype <- ggplot(res_index, aes(x=as.factor(Group), y=EpiTraceAge_iterative, fill = Group)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_EpiScorebyCelltypeQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=EpiTraceAge_iterative, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_CloneLevelByCellGroup <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(cell_group), y = scMutScore)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "scMuttraceScore") +
  theme_classic()

p_CloneLevelByDaysSplitbyCellgroup <- ggplot(res_index_meta_summary_scMutScore, aes(x=factor(cell_day), y=scMutScore)) +
  geom_line(aes(group=clone), size=0.5, alpha=0.1, linetype='dashed') + geom_boxplot(outlier.alpha = 1, fill="white") +
  facet_wrap(~cell_group) + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + theme(text = element_size(12))
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

p_CloneLevelByDaysSplitbyCellType <- ggplot(res_index_meta_summary_scMutScore, aes(x=factor(cell_day), y=scMutScore)) +
  geom_line(aes(group=clone), size=0.5, alpha=0.1, linetype='dashed') + geom_boxplot(outlier.alpha = 1, fill="white") +
  facet_wrap(~Group) + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + theme(text = element_size(12))

p_CloneLevelByDays <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(CultureDay), y = scMutScore)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = day_color) +
  labs(x = "", y = "scMuttraceScore") +
  theme_classic()

```

```

# EpiTraceAge
res_index %>% dplyr::group_by(clone,Group,cell_day) %>% dplyr::summarise(meanEpi=mean(EpiTraceAge_itera
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `.groups` argument.
res_index_meta_summary_EpiTraceAge$cell_group <- 'Prog'
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my1',res_index_meta_summary_EpiTraceAge$Group)] <-
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my2|my3|my4',res_index_meta_summary_EpiTraceAge$Gr
res_index_meta_summary_EpiTraceAge$cell_group <- factor(res_index_meta_summary_EpiTraceAge$cell_group,1

p_CloneLevelByCellGroup_EpiTraceAge <- ggplot(res_index_meta_summary_EpiTraceAge, mapping = aes(x = as.
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_CloneLevelByCellGroup_scMutraceScore <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = a
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_pseudotimeByDaysEry <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudotime_ery, fill = CellMut
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_pseudotimeByDaysMy <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudotime_my, fill = CellMutT
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_pseudotimeByGroupQunEry <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudotime_ery, fill = Group
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_pseudotimeByGroupQunMy <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudotime_my, fill = GroupQ
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_UMAP_scMutrace <- ggplot(data = res_index, mapping = aes(x = x, y = y, color = scMutScore)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option="rocket") +
  xlab("") + ylab("") +
  theme_classic()

df_plot <- res_index %>%
  count(CellMutTag, GroupQun) %>%
  complete(
    CellMutTag,
    GroupQun = c(
      "prog", "prog_my",
      "my1", "my2", "my3", "my4",

```



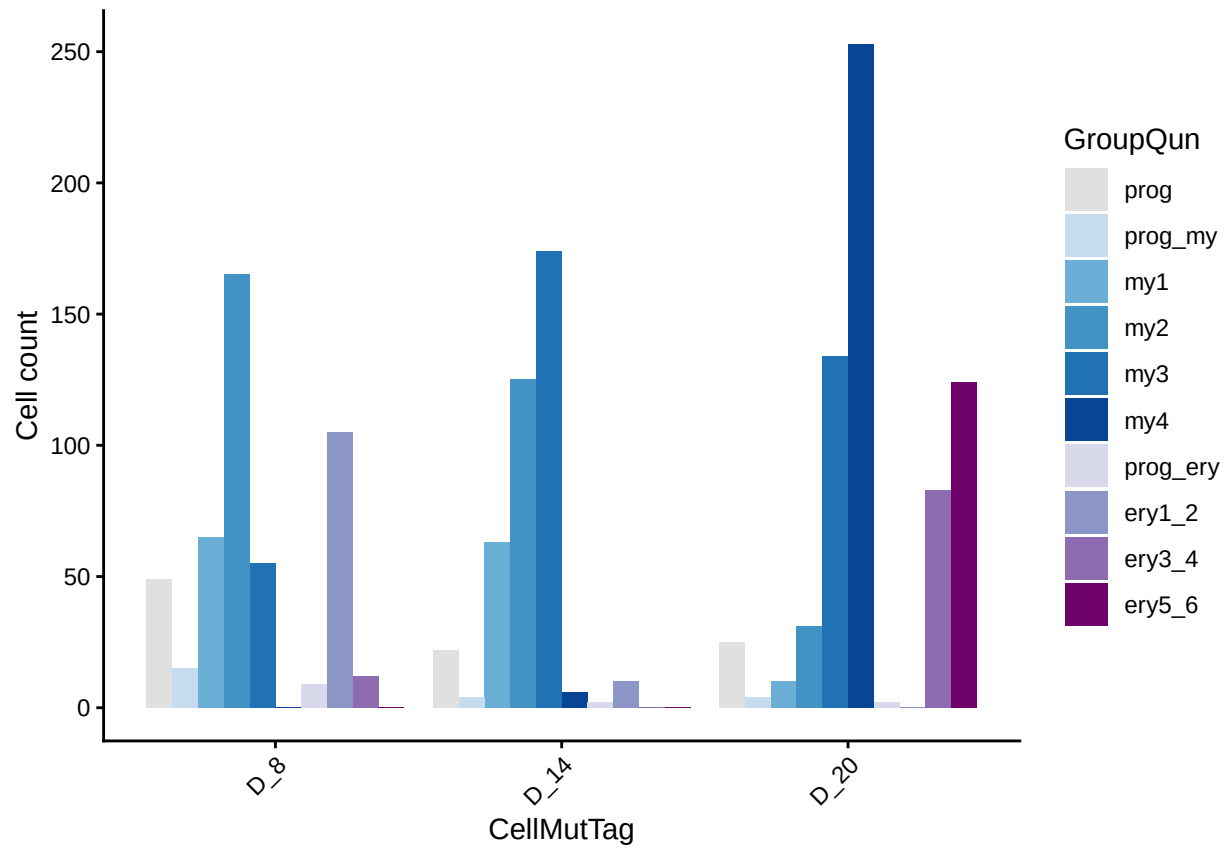
```

      "prog_ery", "ery1_2", "ery3_4", "ery5_6"
    ),
    fill = list(n = 0)
  )

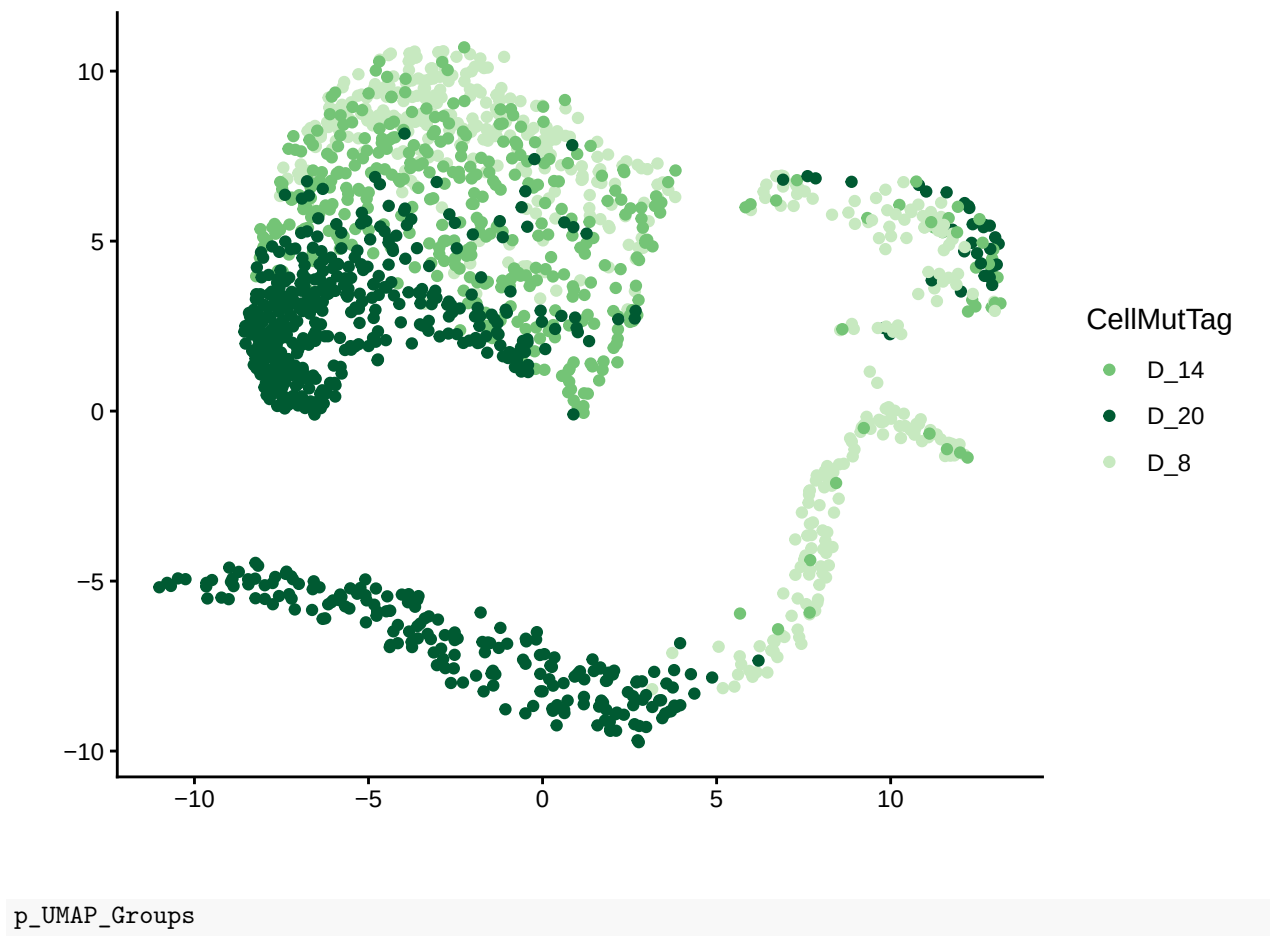
df_plot$GroupQun <- factor(
  df_plot$GroupQun,
  levels = c(
    "prog", "prog_my",
    "my1", "my2", "my3", "my4",
    "prog_ery", "ery1_2", "ery3_4", "ery5_6"
  )
)

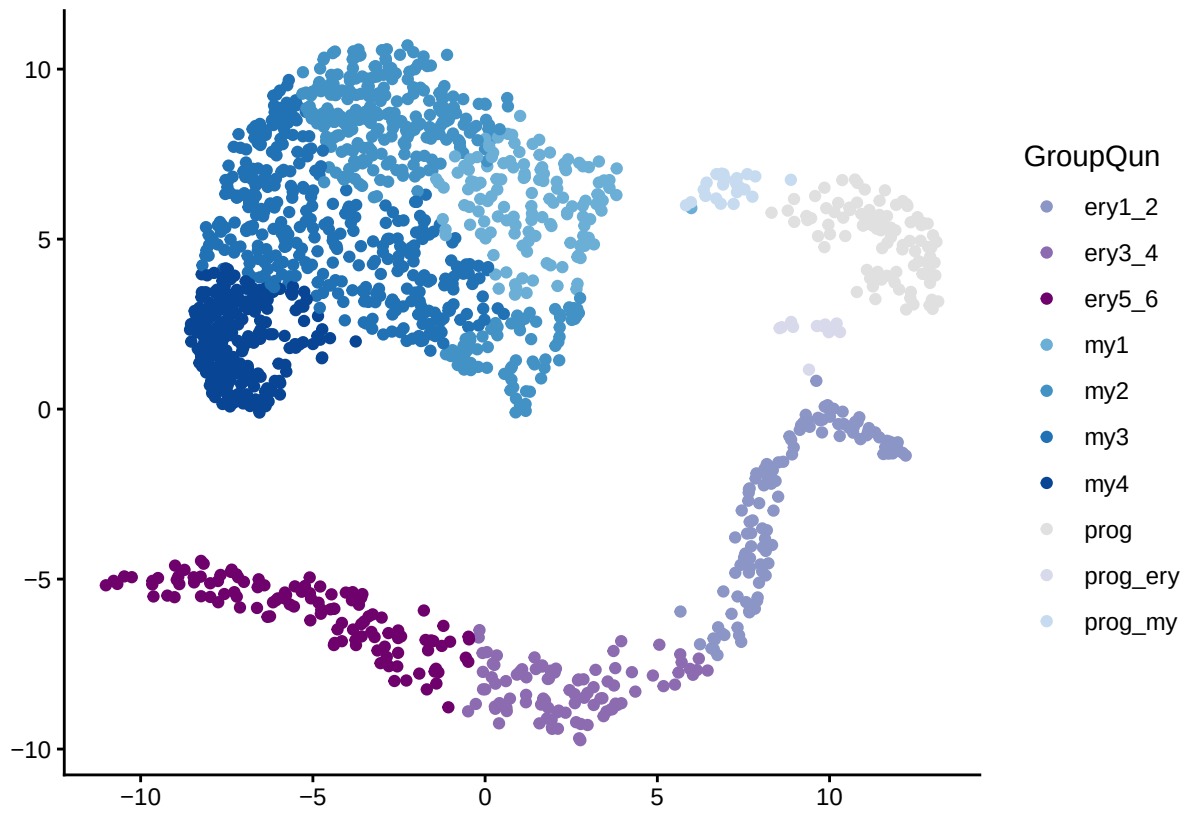
ggplot(df_plot, aes(x = CellMutTag, y = n, fill = GroupQun)) +
  geom_col(position = "dodge") +
  labs(
    x = "CellMutTag",
    y = "Cell count",
    fill = "GroupQun"
  ) +
  scale_fill_manual(values = cluster_color) +
  theme_classic() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

```

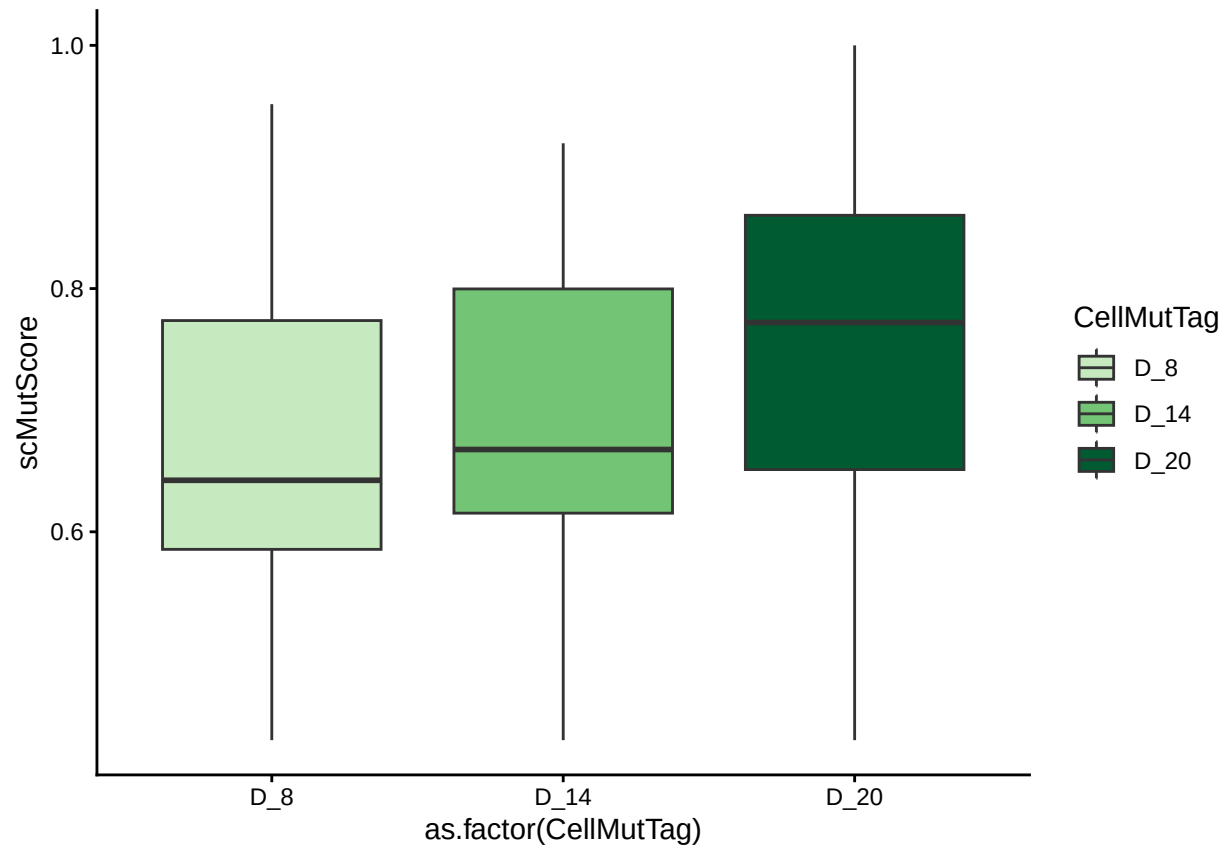


p_UMAP_Days

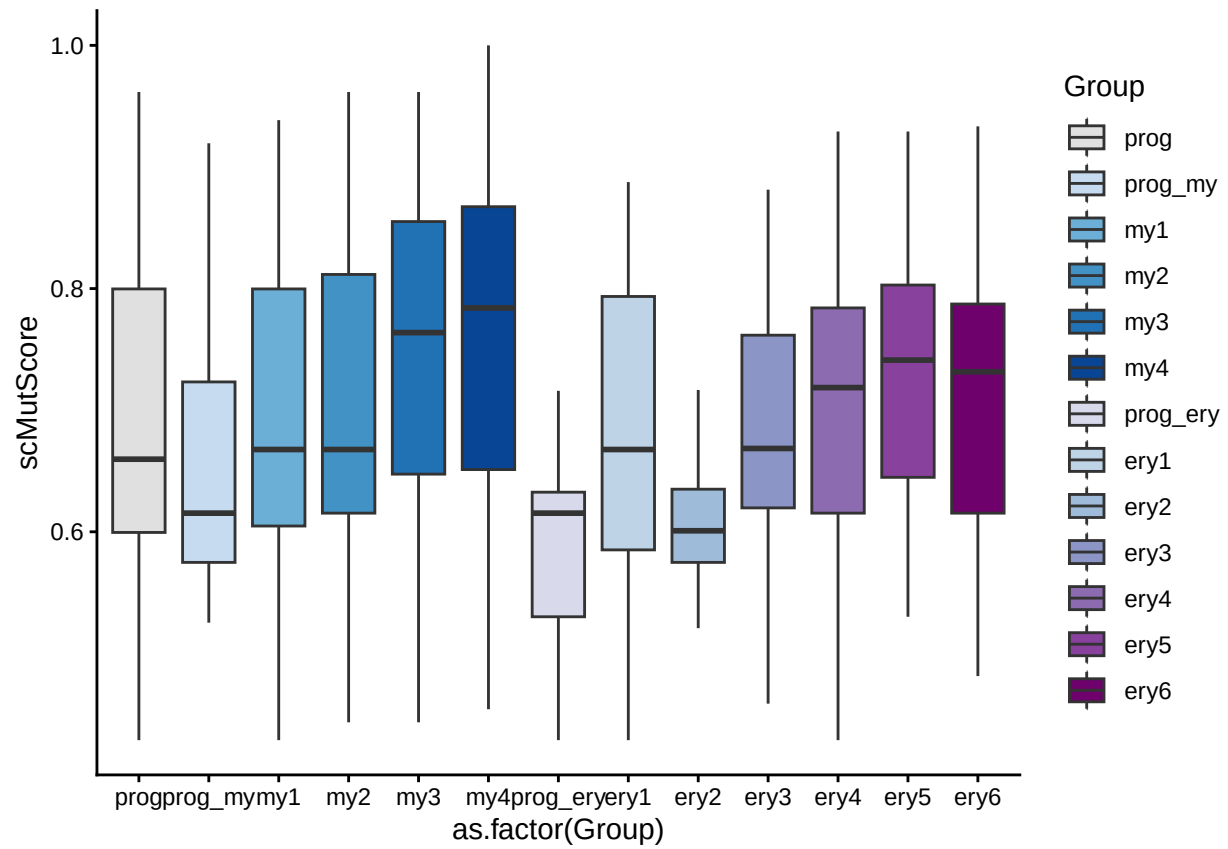




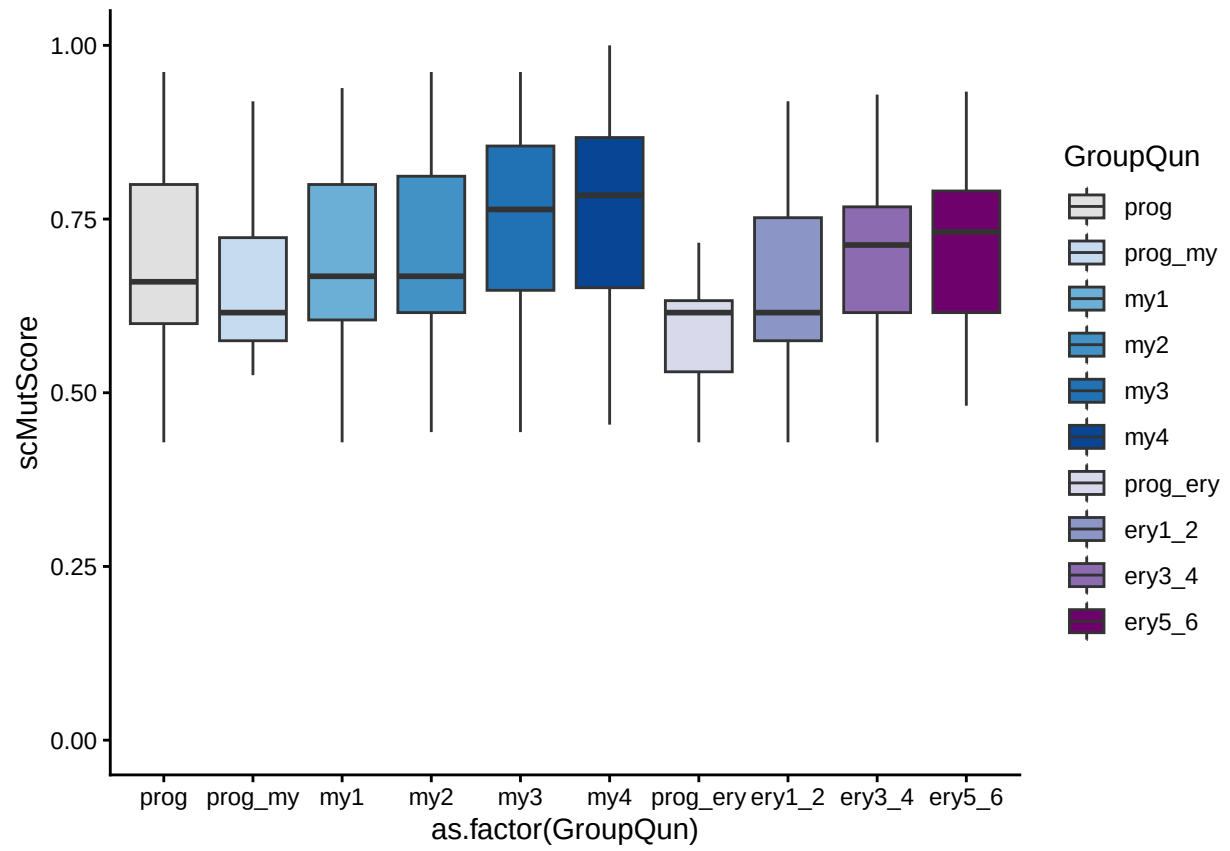
p_scMutScorebydays



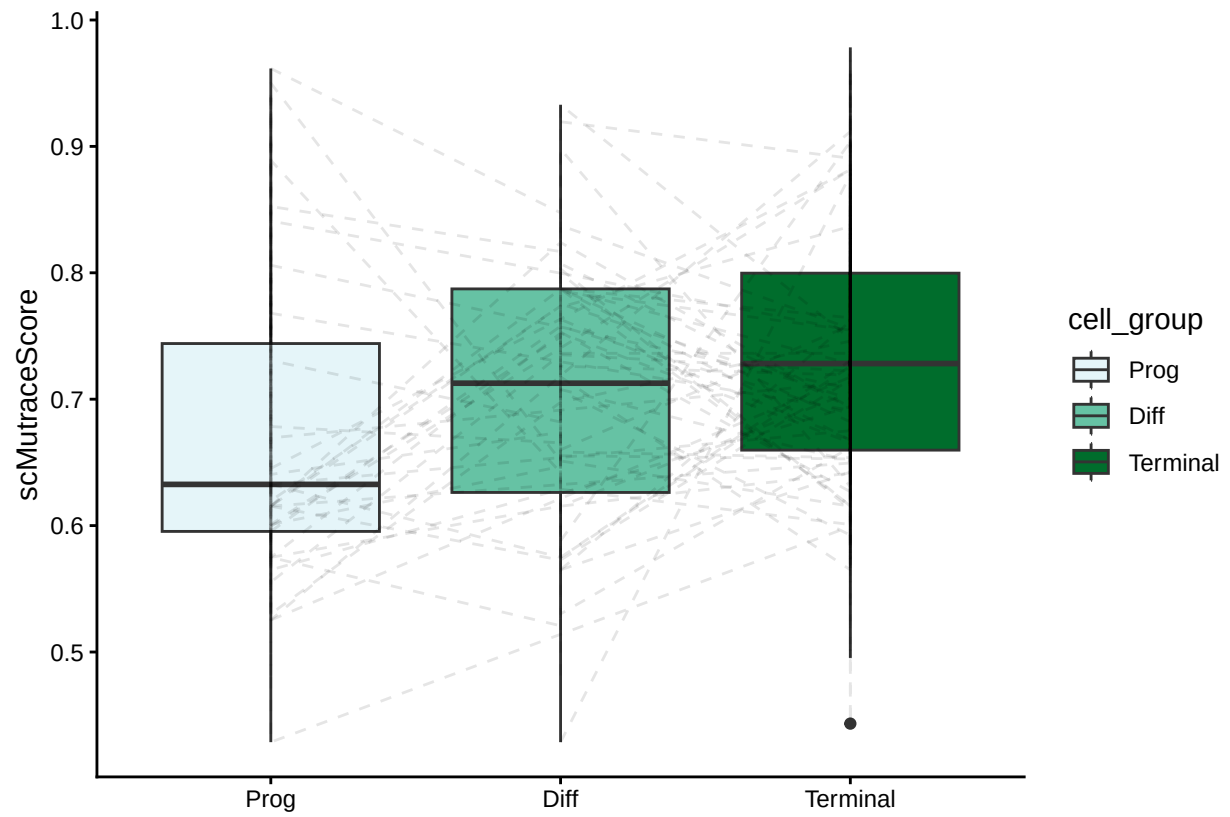
p_scMutScorebycelltypes



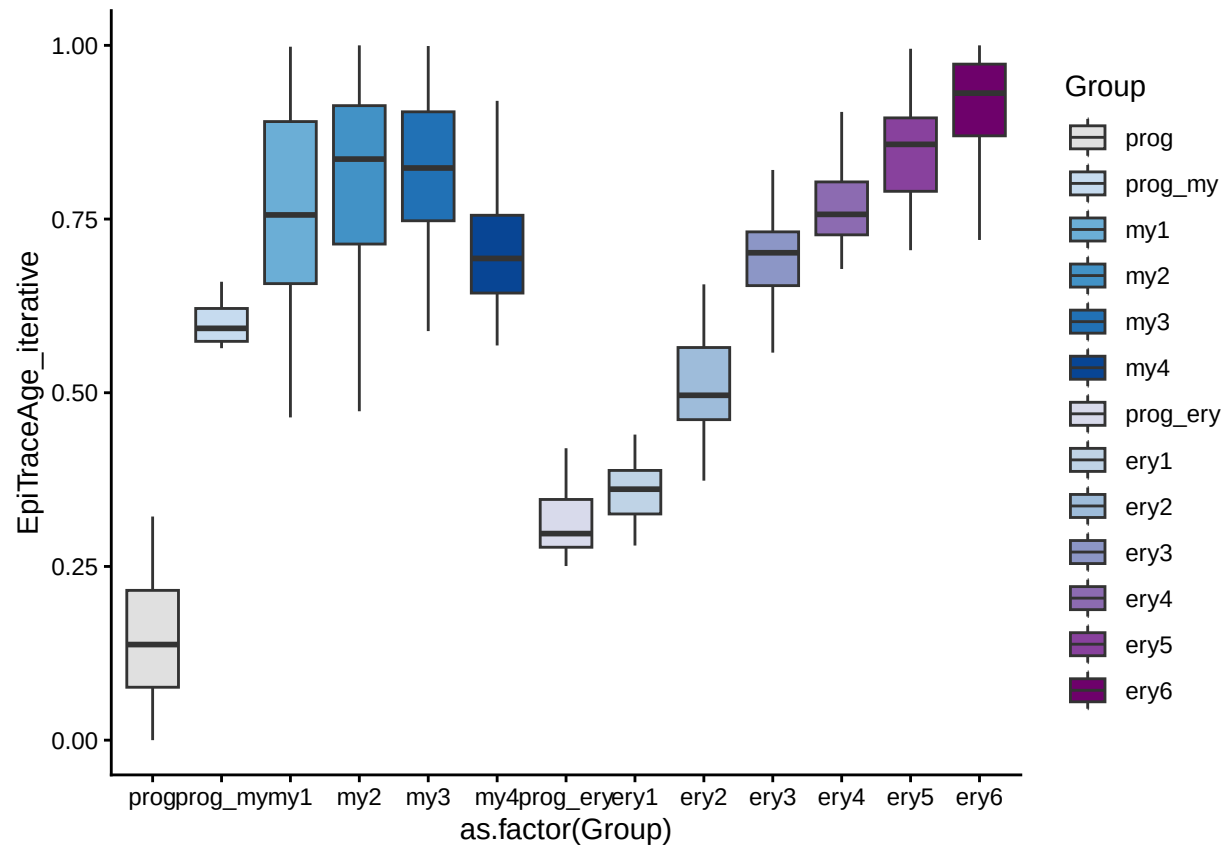
p_scMutScorebycelltypesQun



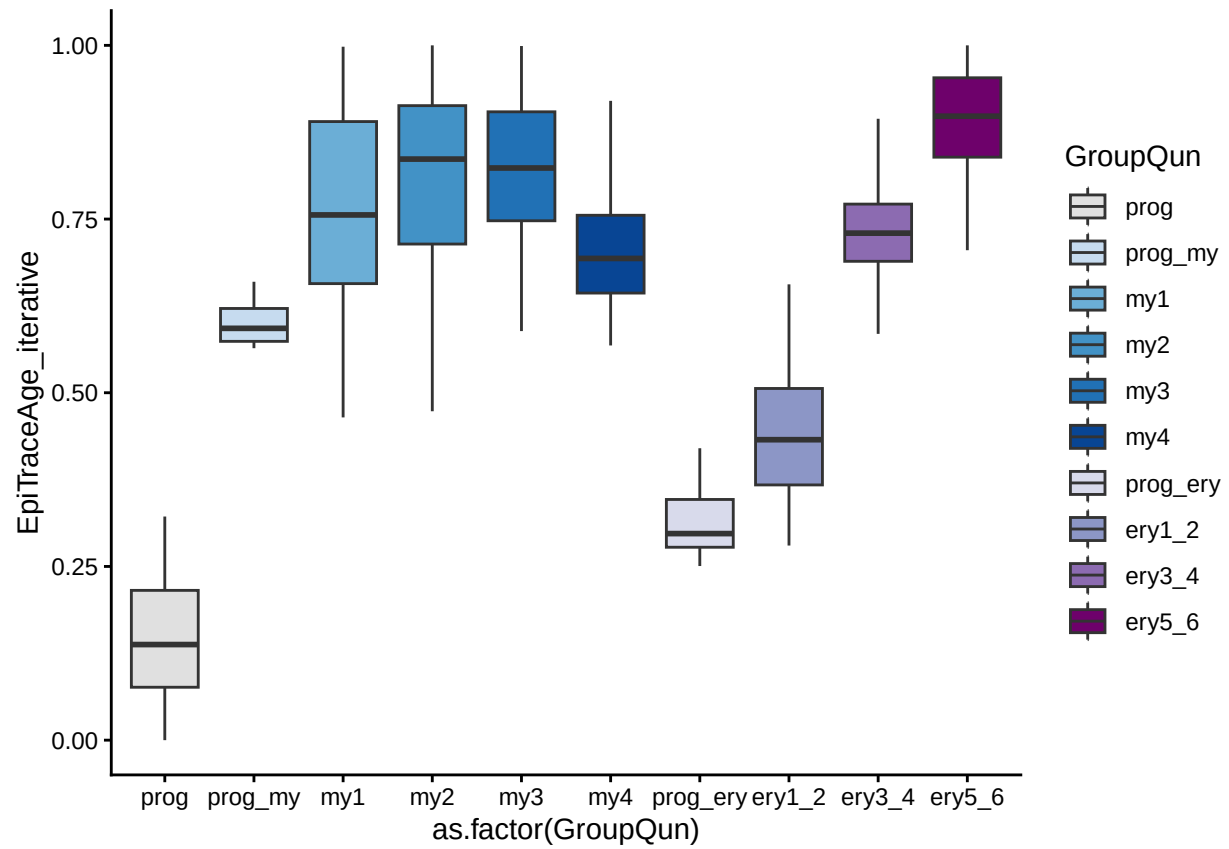
p_CloneLevelByCellGroup



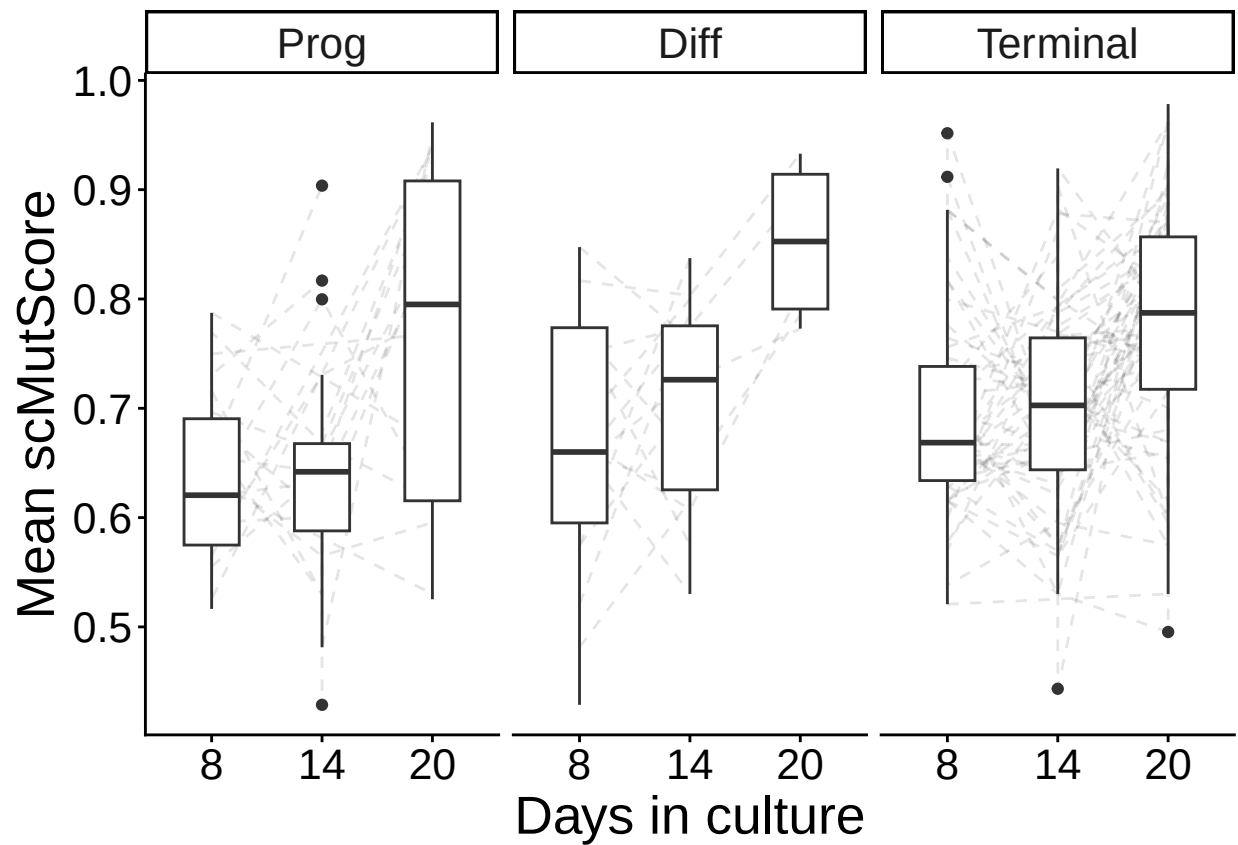
p_EpiScorebyCelltype



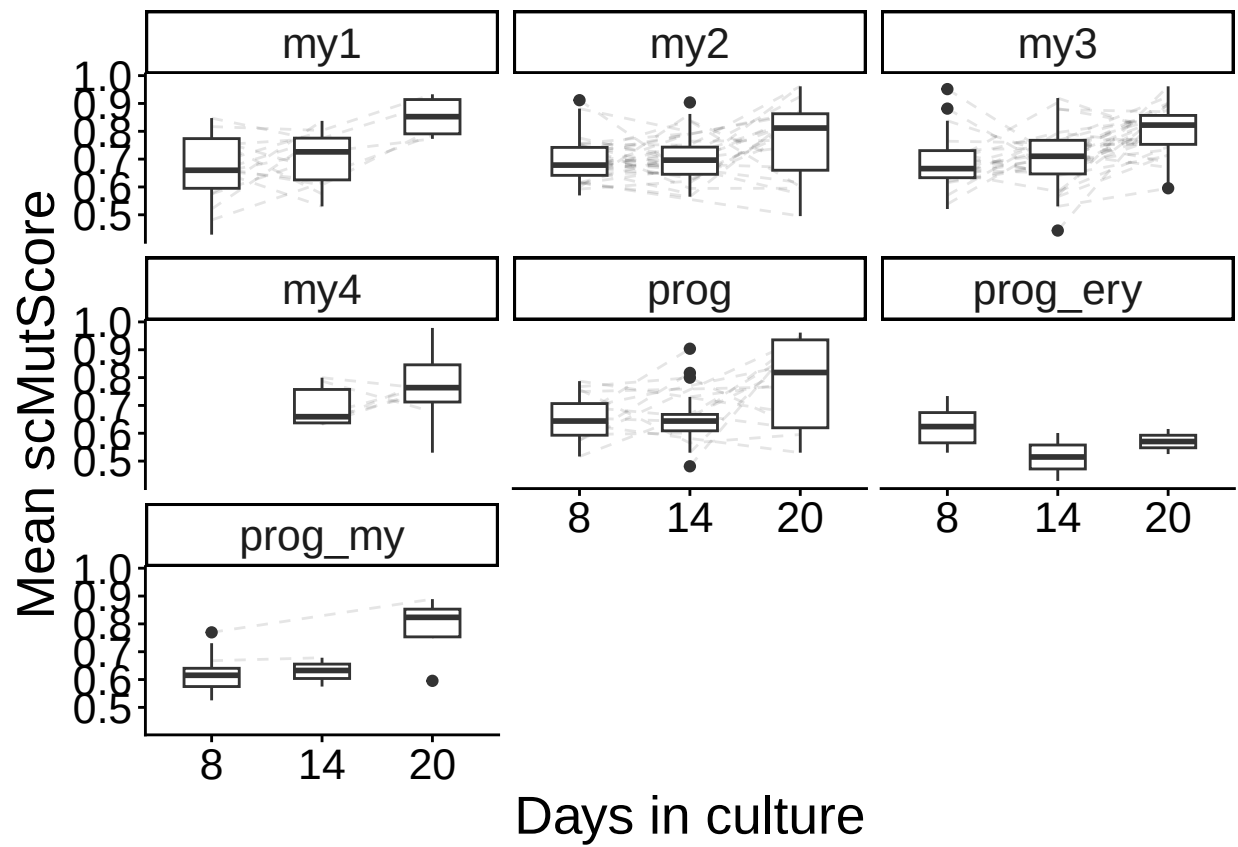
p_EpiScorebyCelltypeQun



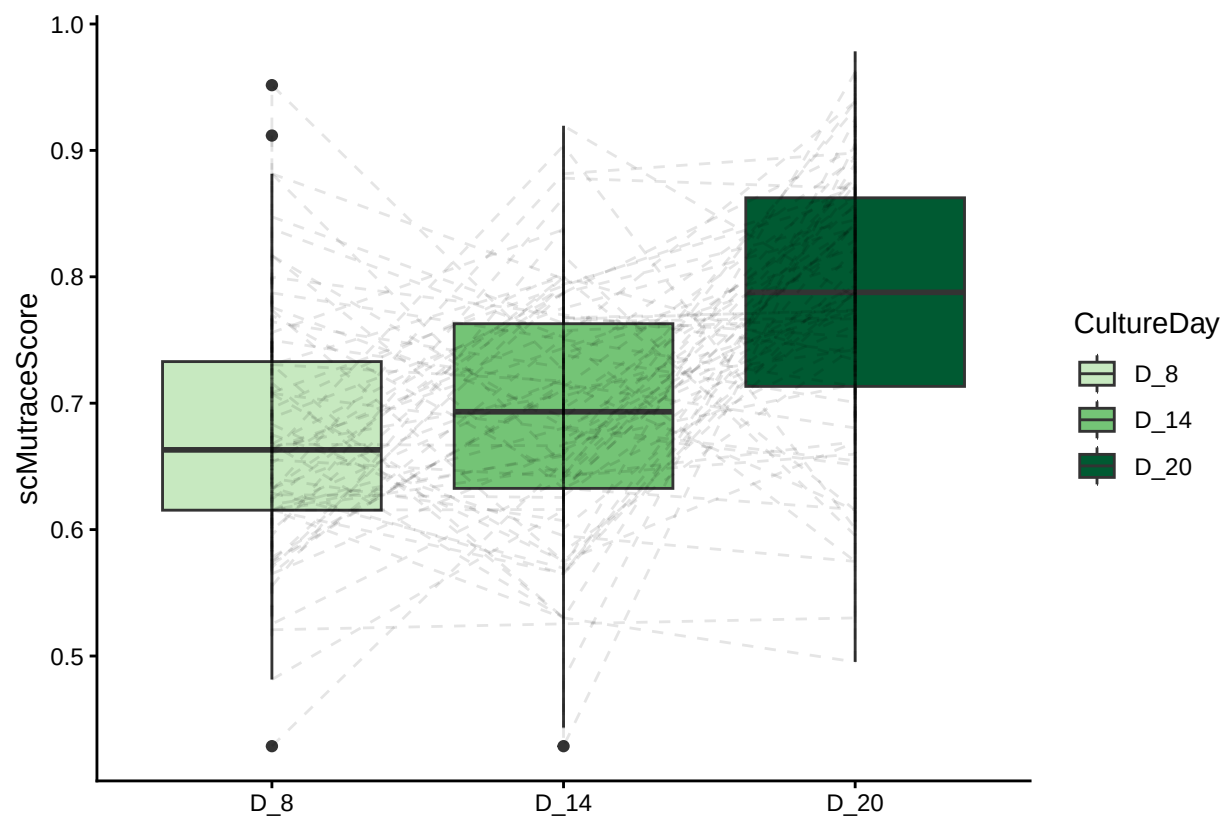
p_CloneLevelByDaysSplitbyCellgroup



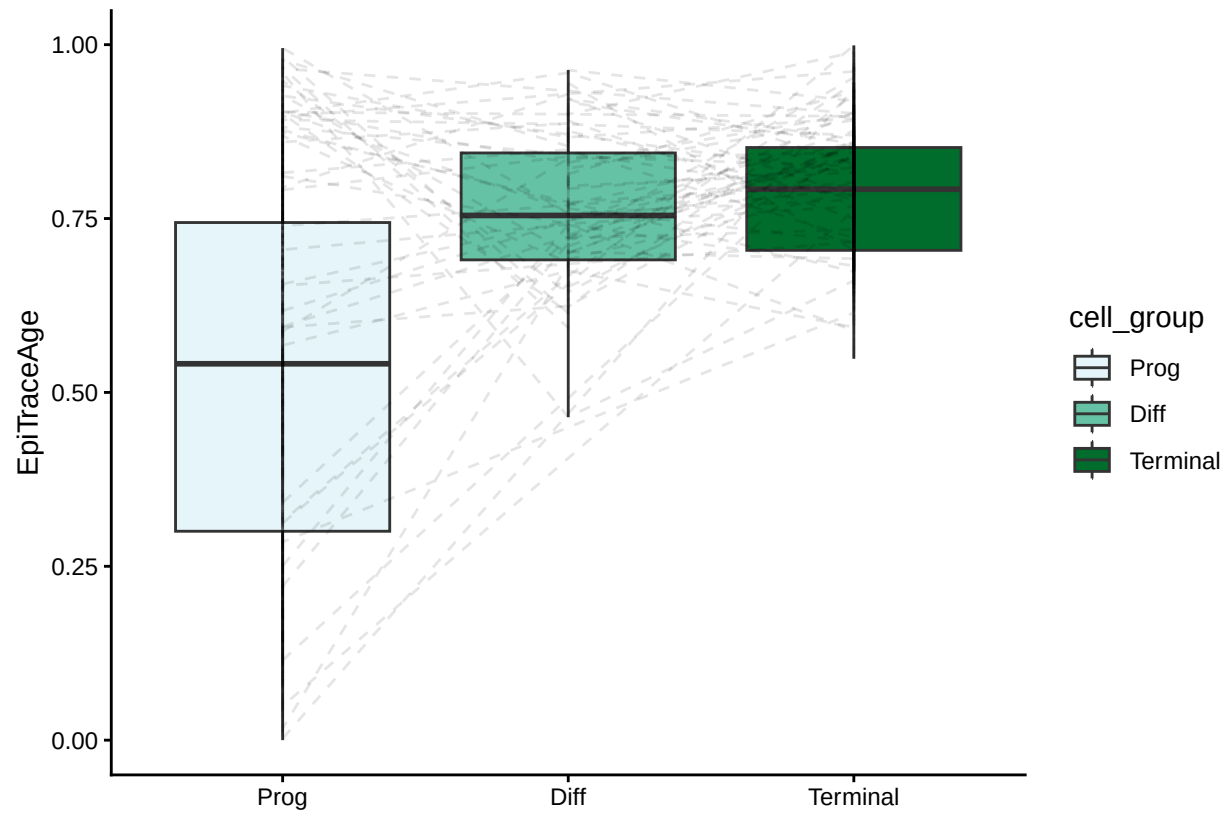
```
p_CloneLevelByDaysSplitbyCellType
## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```



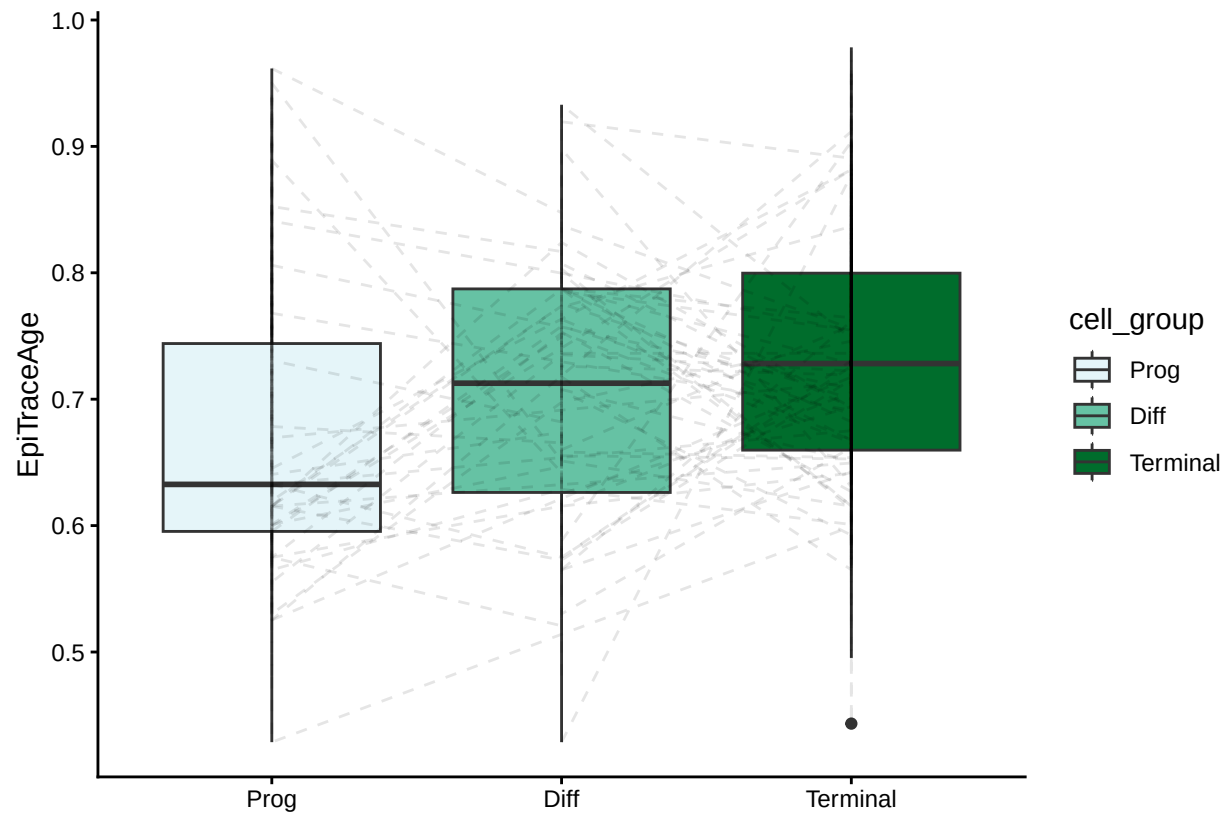
p_CloneLevelByDays



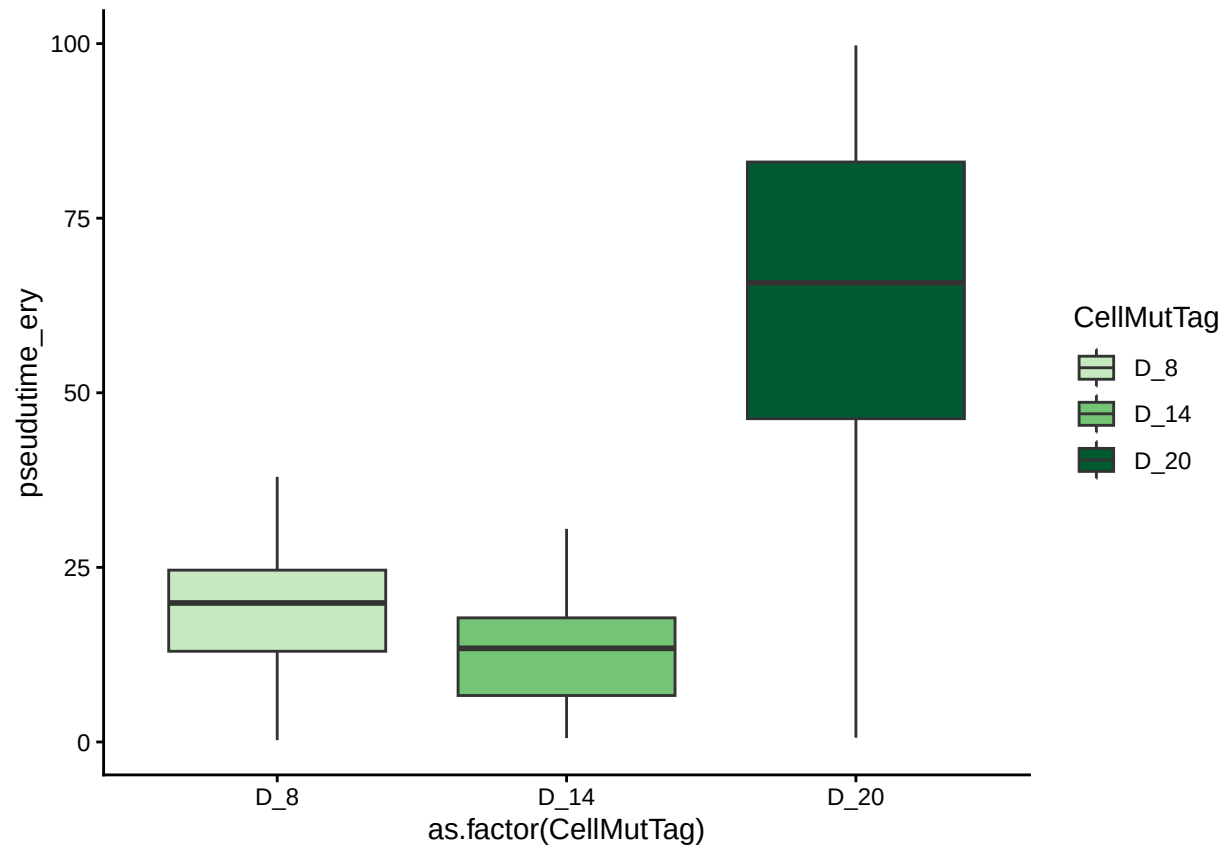
p_CloneLevelByCellGroup_EpiTraceAge



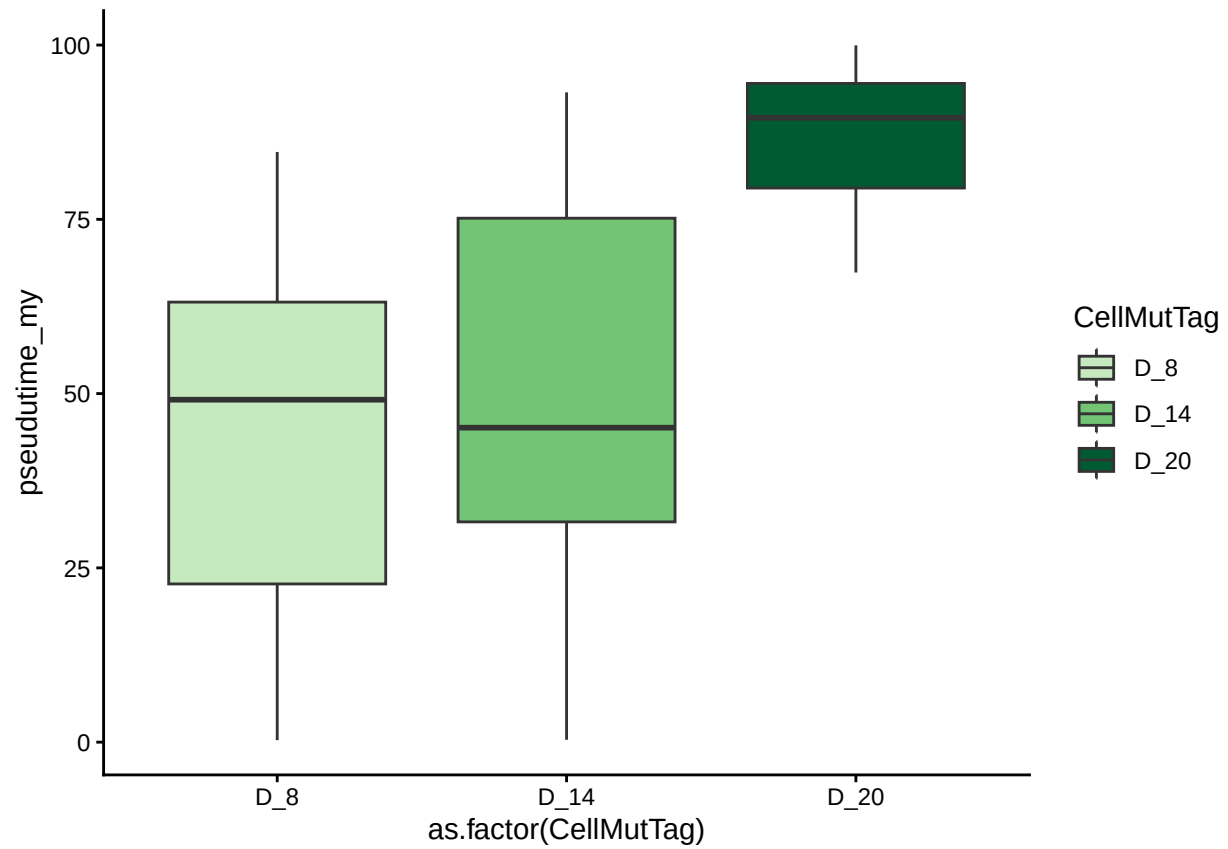
p_CloneLevelByCellGroup_scMutraceScore



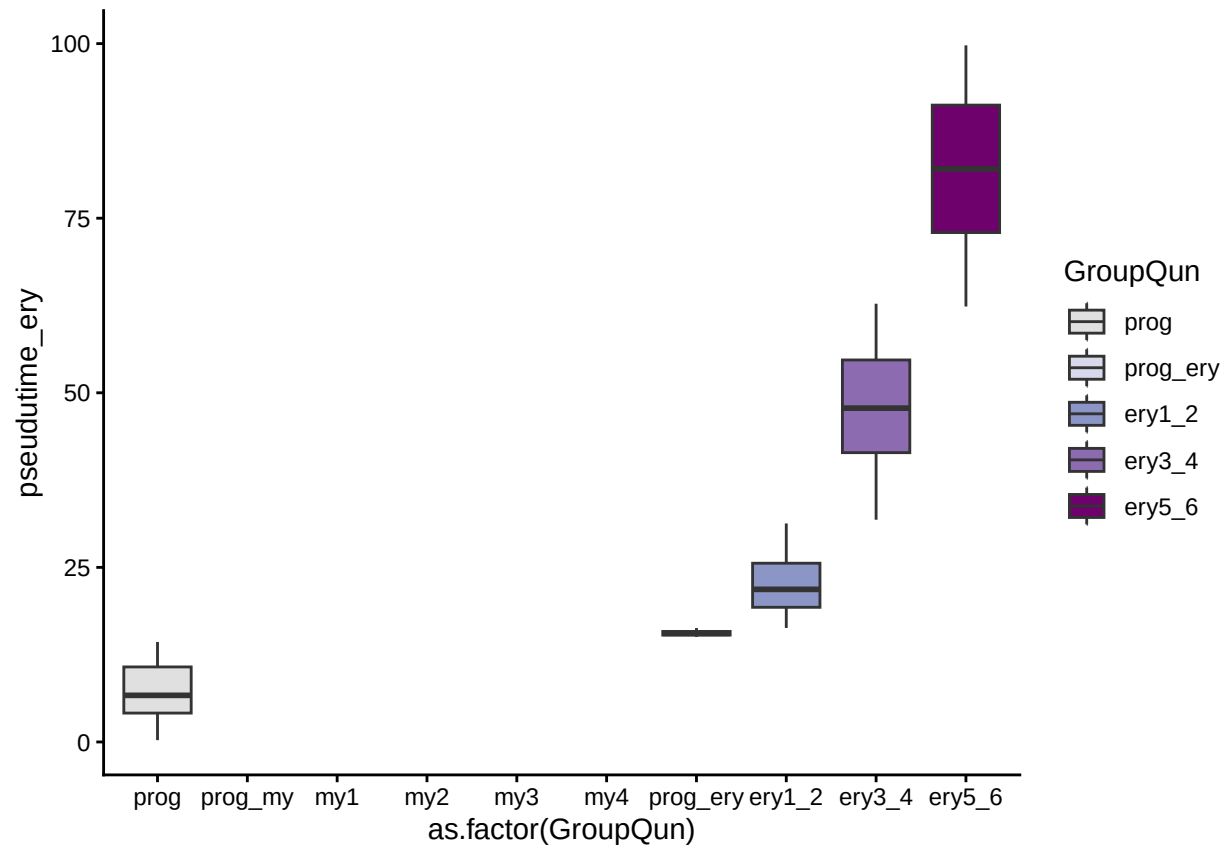
```
p_pseudotimeByDaysEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



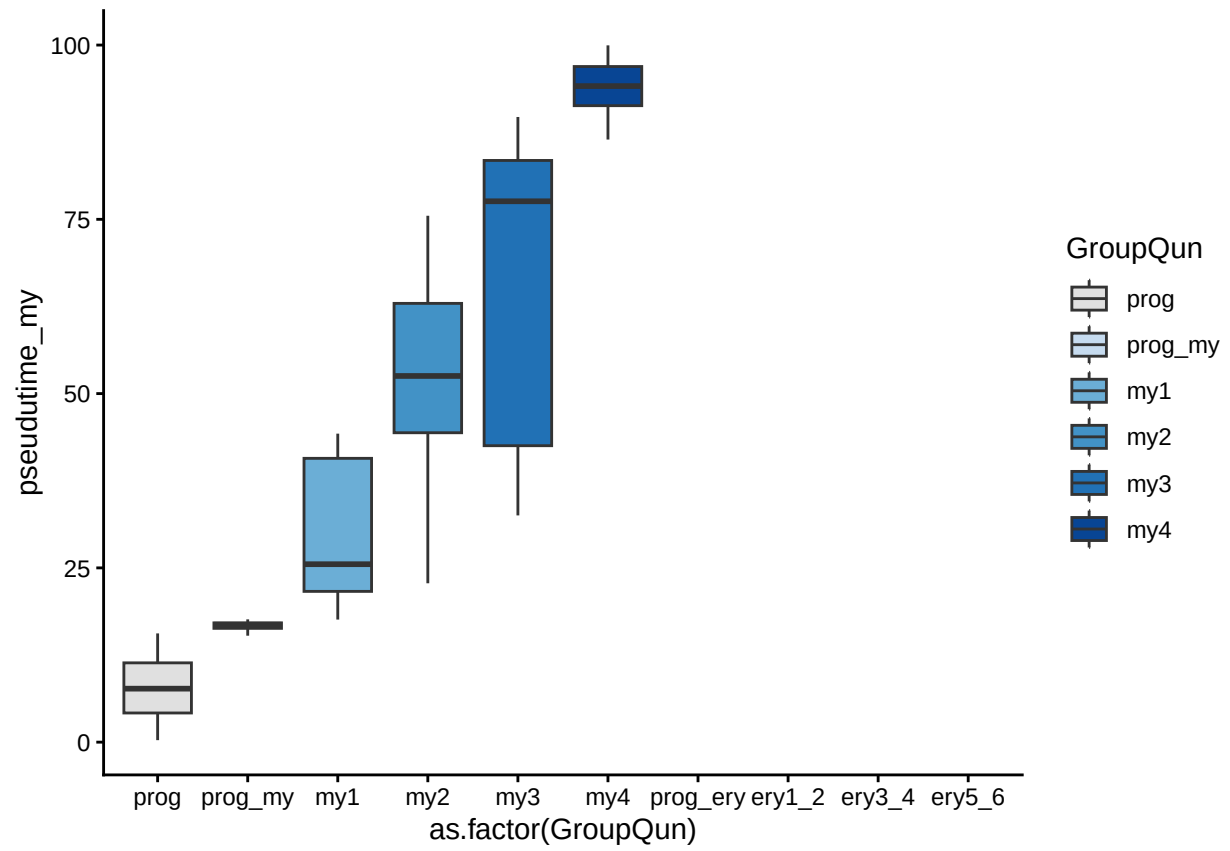
```
p_pseudotimeByDaysMy  
## Warning: Removed 347 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```

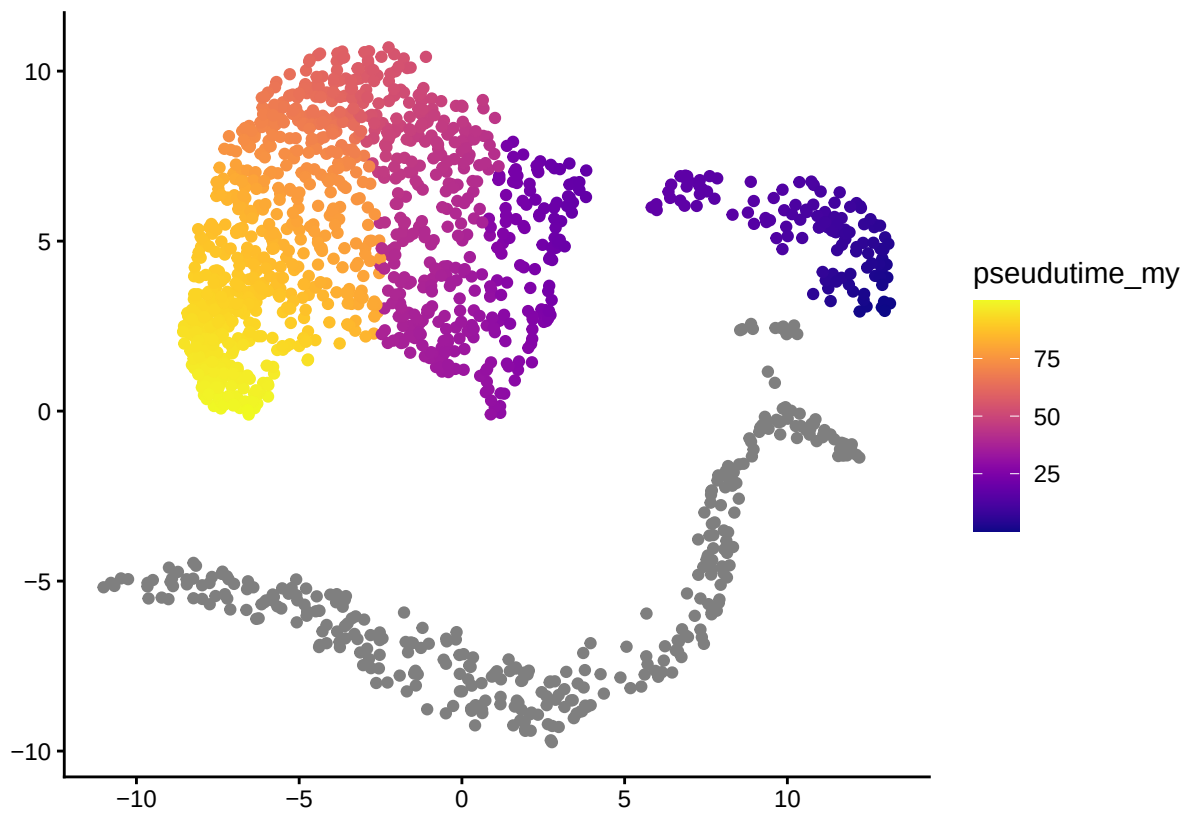
```
p_pseudotimeByGroupQunEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



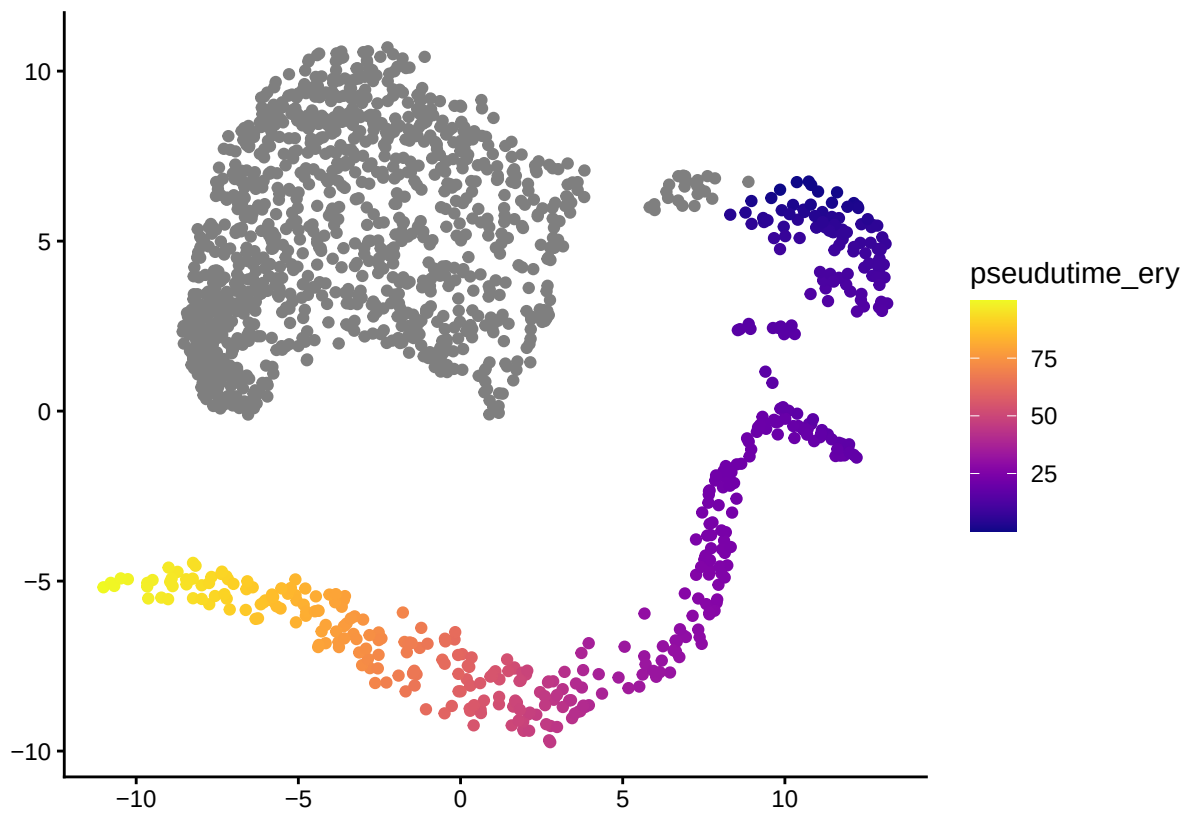
```
p_pseudotimeByGroupQunMy
## Warning: Removed 347 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



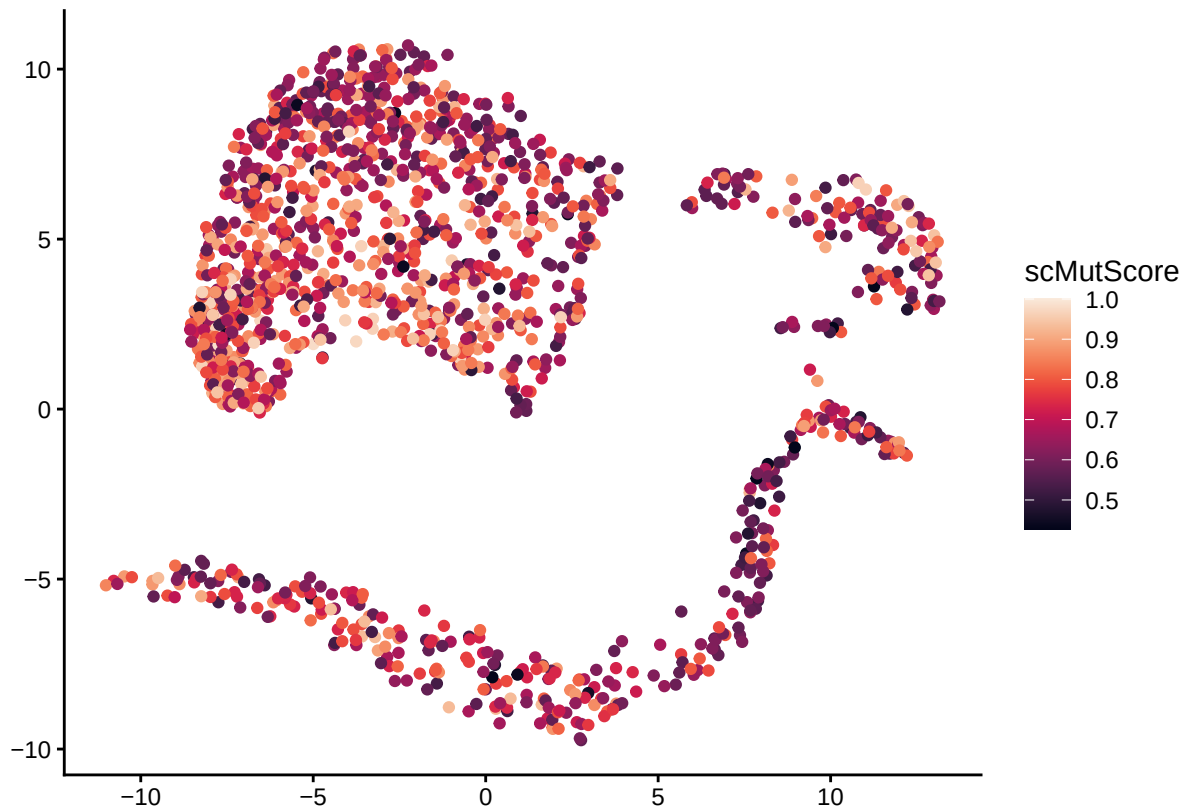
p_UMAP_pseudotime_my



p_UMAP_pseudotime_ery



p_UMAP_scMutrace



8 savePic

```
pdf("../Figure/CD34/p_UMAP_Days.pdf", width = 4.47, height = 3.20)
p_UMAP_Days
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_UMAP_Groups.pdf", width = 4.47, height = 3.20)
p_UMAP_Groups
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_UMAP_pseudotime_my.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudotime_my
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_UMAP_pseudotime_ery.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudotime_ery
dev.off()
```

```

## pdf
## 2

pdf("../Figure/CD34/p_scMutScorebydays.pdf", width = 4.47, height = 3.20)
p_scMutScorebydays
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_scMutScorebycelltypes.pdf", width = 4.47, height = 3.20)
p_scMutScorebycelltypesQun
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_EpiScorebyCelltype.pdf", width = 4.47, height = 3.20)
p_EpiScorebyCelltypeQun
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_pseudotimeByDaysEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByDaysEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_pseudotimeByDaysMy.pdf", width = 4.47, height = 3.20)
p_pseudotimeByDaysMy
## Warning: Removed 347 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_pseudotimeByGroupQunEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
dev.off()
## pdf
## 2

pdf("../Figure/CD34/p_pseudotimeByGroupQunMy.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunMy
## Warning: Removed 347 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
dev.off()
## pdf
## 2

```

9 sessionInfo

```
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] patchwork_1.3.2    lubridate_1.9.4    forcats_1.0.0      purrr_1.1.0
## [5] readr_2.1.5        tibble_3.3.0       tidyverse_2.0.0    networkD3_0.4.1
## [9] tidygraph_1.3.1    reshape2_1.4.4     pheatmap_1.0.13    corrplot_0.95
## [13] stringr_1.5.2      ggpmisc_0.6.2      ggpp_0.5.9         webshot_0.5.5
## [17] ggraph_2.2.2       igraph_2.1.4       Seurat_5.3.0       SeuratObject_5.2.0
## [21] sp_2.2-0           ggpubr_0.6.1       ggplot2_4.0.0      tidyr_1.3.1
## [25] dplyr_1.1.4        vegan_2.7-1        permute_0.9-8      ineq_0.2-13
## [29] FSA_0.10.0
##
## loaded via a namespace (and not attached):
## [1] RcppAnnoy_0.0.22    splines_4.5.1      later_1.4.4
## [4] polyclip_1.10-7     fastDummies_1.7.5  lifecycle_1.0.4
## [7] rstatix_0.7.2       globals_0.18.0     lattice_0.22-7
## [10] MASS_7.3-65         backports_1.5.0    magrittr_2.0.4
## [13] plotly_4.11.0       rmarkdown_2.29     yaml_2.3.10
## [16] httpuv_1.6.16       sctransform_0.4.2  spam_2.11-1
## [19] spatstat.sparse_3.1-0 reticulate_1.43.0  cowplot_1.2.0
## [22] pbapply_1.7-4       RColorBrewer_1.1-3 abind_1.4-8
## [25] Rtsne_0.17          tweenr_2.0.3       data.tree_1.2.0
## [28] ggrepel_0.9.6       irlba_2.3.5.1      listenv_0.9.1
## [31] spatstat.utils_3.2-0 MatrixModels_0.5-4 goftest_1.2-3
## [34] RSpectra_0.16-2     spatstat.random_3.4-2 fitdistrplus_1.2-4
## [37] parallelly_1.45.1   codetools_0.2-20   ggforce_0.5.0
## [40] tidyselect_1.2.1     farver_2.1.2       viridis_0.6.5
## [43] matrixStats_1.5.0    spatstat.explore_3.5-3 jsonlite_2.0.0
## [46] progressr_0.16.0     Formula_1.2-5      ggribes_0.5.7
## [49] survival_3.8-3      tools_4.5.1        ica_1.0-3
## [52] Rcpp_1.1.0          glue_1.8.0         gridExtra_2.3
## [55] xfun_0.53           mgcv_1.9-3         withr_3.0.2
## [58] fastmap_1.2.0       SparseM_1.84-2     digest_0.6.37
## [61] timechange_0.3.0     R6_2.6.1           mime_0.13
## [64] colorspace_2.1-2     scattermore_1.2    tensor_1.5.1
```



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## [67] dichromat_2.0-0.1      spatstat.data_3.1-8    generics_0.1.4
## [70] data.table_1.17.8      graphlayouts_1.2.2    httr_1.4.7
## [73] htmlwidgets_1.6.4      uwot_0.2.3            pkgconfig_2.0.3
## [76] gtable_0.3.6           lmtest_0.9-40         S7_0.2.0
## [79] htmltools_0.5.8.1      carData_3.0-5         dotCall64_1.2
## [82] scales_1.4.0           png_0.1-8             spatstat.univar_3.1-4
## [85] knitr_1.50             rstudioapi_0.17.1     tzdb_0.5.0
## [88] nlme_3.1-168           zoo_1.8-14            cachem_1.1.0
## [91] KernSmooth_2.23-26     parallel_4.5.1        miniUI_0.1.2
## [94] pillar_1.11.1          grid_4.5.1            vctrs_0.6.5
## [97] RANN_2.6.2             promises_1.3.3         car_3.1-3
## [100] xtable_1.8-4           cluster_2.1.8.1        evaluate_1.0.5
## [103] cli_3.6.5             compiler_4.5.1         crayon_1.5.3
## [106] rlang_1.1.6           future.apply_1.20.0    ggsignif_0.6.4
## [109] labeling_0.4.3         plyr_1.8.9            stringi_1.8.7
## [112] viridisLite_0.4.2     deldir_2.0-4          lazyeval_0.2.2
## [115] spatstat.geom_3.6-0    quantreg_6.1          Matrix_1.7-4
## [118] RcppHNSW_0.6.0        hms_1.1.3             future_1.67.0
## [121] shiny_1.11.1          ROCR_1.0-11           broom_1.0.10
## [124] memoise_2.0.1         polynom_1.4-1

```